OpenURMA: A Clean-Room Open Implementation of the Unified Bus Protocol

Bojie Li[1]

[1]Pine AI , bojieli@gmail.com

Modern datacenter RDMA is bottlenecked at the network interface, not at the wire. At the scale of AI-training collectives and HPC all-to-alls, a NIC running RoCE or InfiniBand must hold per-connection state for every (application, remote-endpoint) pair — hundreds of megabytes of context at 1024-application fanout — and pays a four-traversal PCIe round trip on the smallest 64-byte operation, inflating end-to-end latency by an order of magnitude beyond the wire delay. Both costs are structural consequences of the Queue-Pair-over-PCIe abstraction that commodity RDMA inherits from InfiniBand.

Huawei's *Unified Bus* (UB) is a public 2025 specification that removes both costs by changing the abstraction. Its transport protocol decouples per-application endpoint state from per-host reliable-transport state, so connection context grows additively rather than multiplicatively; it exposes ordering as an opt-in surface so applications pay gating only when they need it; and it reaches remote memory through a CPU's native load/store instructions to a controller on the on-chip bus rather than through verbs to a PCIe-attached device. The protocol ships in Huawei's Ascend 950 silicon but has received almost no open analysis: there is no peer-reviewed measurement, no reference implementation outside Huawei, and no controlled comparison against RoCE on identical infrastructure.

OpenURMA is the first clean-room open implementation of the Unified Bus protocol's transport and transaction layers, written from the public specification. We realise it at three modeling tiers — synthesisable RTL on Alveo U50, a cycle-level two-node SystemC simulator that accounts for both endpoints of the wire, and a gem5 full-system scaffold in which two ARM CPUs run real user binaries against the SystemC NIC — with a matched OpenRoCE (RoCEv2 RC) baseline at every tier. The contribution is not the architecture, which is Huawei's; it is the implementation, the multi-tier evaluation harness, and the controlled comparison that closed silicon does not admit.

The artifact lets the community measure the protocol's costs cell-by-cell, locate its operating points against RoCE on identical infrastructure, and prototype follow-on transports on a common substrate. We use it to show that the three architectural choices UB makes are mutually load-bearing: removing any one and keeping the others on a PCIe-attached NIC reproduces RoCE's costs. On a 64-byte READ, the load/store path delivers ≈500 ns end-to-end CPU-to-remote-DRAM-to-CPU latency, **4.37**× below the matched RoCEv2 baseline (2186 ns), at roughly 14% of a U50's LUT budget. We release OpenURMA so the rest of the questions Unified Bus raises become investigable outside the closed silicon that currently ships.

1. Introduction

1.1 RDMA is bottlenecked at the NIC, not the wire

The connection-cardinality of modern datacenter workloads has overtaken link bandwidth as the binding resource of an RDMA NIC. An AI-training job that exchanges activations all-to-all across N co-located applications and M remote endpoints opens roughly $N{\cdot}M$ sender–receiver pairs. Under RoCEv2 with Reliable Connection (RC) — the dominant production RDMA mode — each pair is a Queue Pair (QP) carrying ∼512 B of NIC state. At $N{=}M{=}1024$, a single NIC's working set is ∼537 MB. That figure does not fit in any commodity NIC's on-chip SRAM, and the moment it spills to host DRAM every operation pays a fresh PCIe round trip to refetch its QP [20, 15, 17, 42].

The same operation pays a second cost even when state stays on-chip. A 64-byte READ on RoCE — the smallest verb the protocol exposes — spends roughly 1.5 μs of its ∼2 μs round trip on four PCIe traversals: the CPU posts a doorbell over MMIO, the NIC fetches the work-queue entry by DMA, the NIC writes the completion entry by DMA, and the CPU polls until the completion miss resolves through PCIe-coherence. The wire delivers in ∼200 ns; the PCIe boundary between CPU and NIC inflates per-op latency by an order of magnitude.

Both costs are *structural* to the QP-over-PCIe abstraction that commodity RDMA inherits from InfiniBand. The QP fuses application identity with reliable-transport state, so per-pair binding is the unit of accounting. The PCIe attachment forces every operation through doorbell-and-completion messaging in disjoint CPU/NIC address spaces. No amount of caching, batching, or flow-control tuning above the device layer removes them. Removing them requires changing the abstraction.

1.2 Unified Bus: a public spec without a public implementation

Huawei's *Unified Bus* (UB) [12] is a 2025 specification that changes the abstraction. It defines a memory-semantic remote-memory-access transport in which a remote read is the same kind of operation a CPU issues to its local DRAM — a `ld`/`st` to a controller on the on-

chip bus — rather than a verb issued to a device behind PCIe. The same protocol bounds NIC state additively in the application and remote-host counts rather than multiplicatively in their product, and exposes ordering as a per-operation opt-in surface rather than an always-on tax. Huawei's Ascend 950 NPU series [13], disclosed in 2025, is the first silicon shipping UB at commodity scale.

The spec is public. The silicon is buyable. Yet the protocol it implements has received almost no open analysis. The specification runs roughly 700 pages and is not yet covered in the systems literature; the Ascend silicon is closed (no RTL, no cycle-level models, no in-network observability); the user-space verb library Huawei released alongside the spec implements the API but does not exercise the NIC datapath that gives UB its latency and area properties. A researcher who wants to *cite* UB has the spec to cite. A researcher who wants to *measure* UB — compare it against RoCE on identical infrastructure, locate the workloads where memory-semantic RDMA wins or loses, or prototype a follow-on transport on top of it — has nothing.

1.3 The three architectural choices UB makes

UB's transport protocol makes three architectural commitments. We describe each at concept altitude here; the spec-level mechanisms, the trade-offs each commitment buys and pays, and our RTL realisation are deferred to §2 and §3.

1. **Decouple application identity from transport reliability.** UB splits transport-layer state into two abstractions: one per local application, holding only the work-queue handles and permissions that identify *who* is calling; and one per remote host, holding the sequence numbers, retransmit queue, and congestion-control state that implement *how* reliable delivery works. Any application can address any remote endpoint through the shared per-host channel, with the destination carried in the packet header rather than baked into a per-pair binding. Per-NIC state grows as $O(N+M)$ instead of $O(N \cdot M)$. The cost the protocol pays for this is one indirection on every transmitted packet to look up the right per-host channel, and a protocol surface that exposes application-side and transport-side resources as separate objects.
2. **Make ordering an opt-in surface.** UB exposes ordering along four orthogonal axes: a per-channel service mode that controls how aggressively packets may reorder on the wire, a per-operation execution tag that controls whether the local issue stage may emit the next operation before the previous one commits, an explicit application-level fence, and a per-operation completion-order bit that controls whether completions appear in arrival order or original issue order. A pure-fanout broadcast bypasses every gating stage; a barrier-after-collective pays exactly the gating its workload needs. RoCE RC, by contrast, enforces strict in-order delivery on every operation regardless of need. The cost UB pays for this flexibility is gating logic that must read the right counters for each mode; the cost it avoids is the strict-order tax on operations that did not need to be ordered in the first place.
3. **Reach remote memory through the on-chip bus, not through PCIe.** For small synchronous operations, UB defines a data path in which the CPU issues bare `ld`/`st` instructions to an address aperture owned by an on-chip controller; the controller translates each into a single-shot network transaction with no work-queue entry, no completion entry, no doorbell, and no DMA. The four PCIe traversals that anchor the per-operation floor of every PCIe-attached RDMA NIC collapse into on-chip-bus crossings roughly an order of magnitude smaller. The cost UB pays is that the controller must sit on the same chip as the CPU it serves; the cost it avoids is the PCIe round trip itself.

These are not three independent optimisations. The on-chip controller of (3) is feasible only because (1) has bounded the NIC's working set to something that fits on the on-chip bus's locality budget; a state table that spills to host DRAM cannot also live on-bus. The opt-in surface of (2) is cheap precisely because (1) has already provisioned the per-application counters that the gating logic indexes against; a NIC without a per-application axis would have to add that machinery from scratch to expose ordering at all. And the multi-path spreading that (1) admits under the unordered service mode is correctness-safe only because the implementation in §3 separates packet-level reordering from operation-level completion ordering. The three are three faces of one architectural commitment, and the central empirical question of this paper is what that commitment actually costs and saves when it meets silicon.

1.4 OpenURMA: the missing open substrate

OpenURMA is a clean-room open-source implementation of the Unified Bus protocol's transport and transaction layers, written from the public specification. The same artifact is observable at three modeling tiers, each chosen because it is load-bearing for a different sub-claim:

- **Synthesisable RTL on Alveo U50.** For the state and area claims: per-element LUT and BRAM, post-route timing closure, the on-chip working-set size that lets the controller live on-bus, and the gating logic's actual cost in pipeline cycles.
- **A cycle-level two-node SystemC simulator.** For per-op latency and sustained throughput: both endpoints of the wire are modeled concurrently, including the work-queue, doorbell, DMA on both sides, the wire link, DRAM, the cache hierarchy, and the completion path. The simulator runs 388 sweep configurations covering all six verb categories.
- **A gem5 full-system scaffold.** For the CPU-to-remote-DRAM end-to-end claim that requires a CPU in the loop: two ARM CPUs run real user binaries against the SystemC NIC linked into gem5 as a memory-mapped device. The PCIe round trip only exists when a CPU exists; this tier is where it actually appears.

A parallel *OpenRoCE* stack implementing RoCEv2 RC on the same toolchain, same target part, and same test

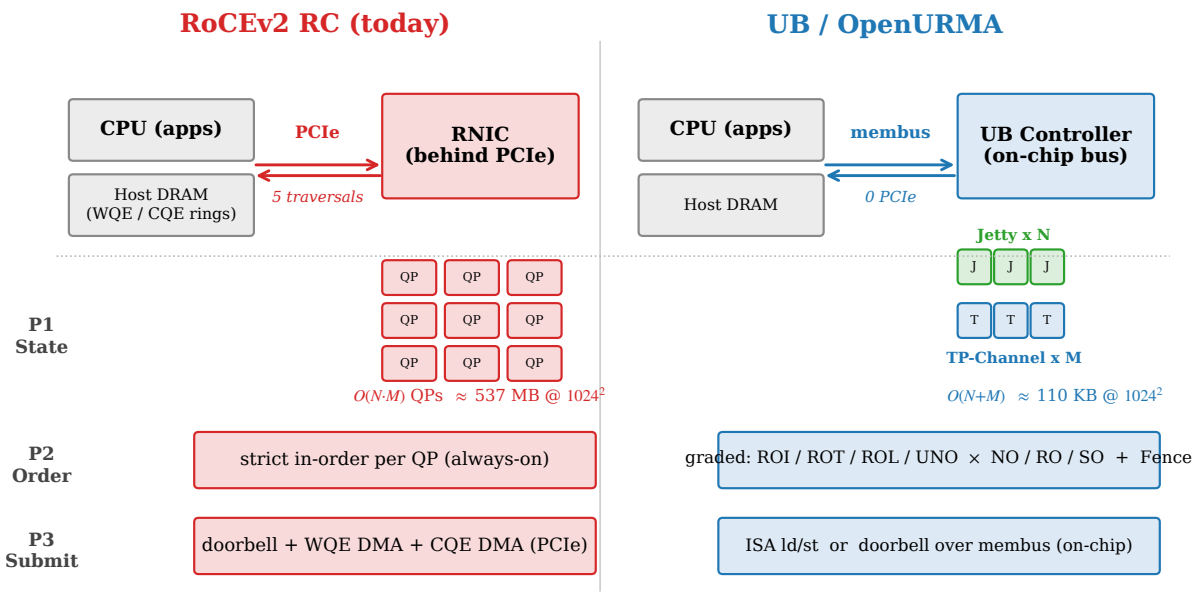


Figure 1: The three architectural choices and how they compose. RoCEv2 RC (top) puts the NIC behind PCIe, holds one Queue Pair per (application, remote-endpoint) pair, and enforces strict order on every operation. UB (bottom) places the controller on the on-chip bus, splits state into per-application and per-host objects, and exposes ordering as four opt-in axes. The arrows mark the couplings the implementation reveals: bounded state is the precondition that lets the controller live on the on-chip bus, the on-chip controller is the precondition that lets a synchronous load/store path exist, and the counters provisioned for the layer split are reused at zero added cost by the ordering surface.

harness anchors an apples-to-apples baseline at every tier. The point of the controlled comparison is to ensure both stacks' numbers come from the same modeling assumptions rather than from published vendor data, which proprietary silicon by construction cannot offer.

The point of releasing the artifact is not to claim architectural novelty — the architecture is Huawei's — nor to match Ascend's silicon in production. The point is that an open implementation makes the questions UB raises tractable for everyone else: how much does the layer split actually cost in LUTs, what does opt-in ordering actually save per operation, where does the PCIe floor end and the wire begin, on which workloads does memory-semantic RDMA win or lose, and how does the architectural commitment generalise to future transports the community is now designing (Falcon, UEC, SRNIC-derivatives). As a concrete preview: on a 64-byte READ in the two-node simulator, the load/store path delivers ≈500 ns end-to-end at **4.37**× below RoCEv2's 2186 ns; how much of that gap survives in production silicon is one of the questions this paper hands to follow-on work.

Contributions.

- **The first clean-room open implementation of the Unified Bus protocol stack.** Synthesisable RTL covering the transport and transaction layers, closing 322 MHz post-route on commodity FPGA at roughly 14% of an Alveo U50's LUT budget.
- **An apples-to-apples OpenRoCE baseline.** A RoCEv2 RC implementation on the same toolchain, same target part, and same test harness, so both stacks' numbers are produced under identical modeling assumptions.
- **Multi-tier evaluation infrastructure.** A cycle-level two-node SystemC simulator that accounts for both endpoints of the wire, plus a gem5 full-system scaffold that places a CPU in the loop — jointly necessary to measure the end-to-end CPU-to-remote-DRAM path the protocol claims to shorten.
- **Ten cross-cutting validation experiments.** Including a model-validation step that reproduces published ConnectX-7 RDMA WRITE latency within ±5%, congestion-control dynamics against DCQCN, and a YCSB-A application port; together these convert UB's analytical claims into measured curves on the open substrate.

OpenURMA is released at `https://github.com/bojieli/OpenURMA`.

2. Background

2.1 UB's Jetty + TP Channel split

A **Jetty** (§8.2.2 of [12]) is an application-level RDMA endpoint identified by a 24-bit Jetty ID. The spec defines three *types* — standard (SQ+RQ), single-sided JFS or JFR, and the target-side *Jetty Group* that aggregates multiple member Jetties under one outward ID and lets the *target NIC* dispatch incoming Sends to a member RQ in hardware (§8.2.2.1 Type 3, dispatch policies: hint-hash, round-robin, RQ-depth). Each Jetty carries a state machine (Reset/Ready/Suspend/Error) with two exception modes, is access-gated by a per-Jetty Token, and supports M2N communication by default: one Jetty addresses any

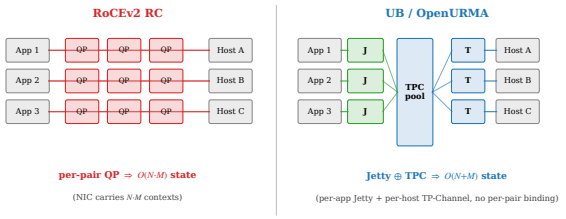


Figure 2: State models compared. RoCE pins one QP to every (application, remote-host) pair; UB decouples per-application state (Jetty) from per-host transport state (TP Channel), and traffic from any Jetty to any host shares a single TP Channel pool with no per-pair binding.

number of remote Jetties without per-pair state, the destination carried in the BTAH header. Per-Jetty state is small (work-queue handles, MR base, JFC handle, state byte) and scales as $O(N)$ in the local app count.

A **TP Channel** (§6.2–6.4) carries the per-remote-host reliable-transport state: PSN window, retransmit queue, in-flight bitmap, RTO timer, cwnd. One TP Channel covers *all* Jetty traffic to one remote host, so the pool scales as $O(M)$. The Initiator multiplexes Jetty traffic onto TP Channels by destination host; the Target demultiplexes by Jetty ID (or Jetty Group, then hardware dispatch) after PSN reordering.

Figure 2 contrasts the two state models. The split has three consequences we will measure separately: (i) **state-byte scaling** $O(N{+}M)$ vs $O(N{\cdot}M)$ — at $(1024, 1024)$, ≈ 110 KB vs ≈ 537 MB (§10.3, Table 2); (ii) **hardware M2N fanout** — one Jetty fans out to any target count through a single TPC pool with no per-pair state and, under Jetty Group, no target-CPU dispatch cost (§8.10); and (iii) **a spec-surface beyond state bytes** — resource ownership (SQ/RQ/JFC), the state machine, and the access-control token are first-class transport-layer objects, not verb-library conventions.

2.2 Graded ordering surface

UB §7.3 exposes ordering as four independent axes rather than a single knob:

- **Service mode (per-channel)**: ROI (Reorder by Initiator), ROT (Reorder by Target), ROL (Reorder by Lower-layer; fuses TPACK with the data-plane ACK), UNO (Unordered).
- **Execution order (per-WR)**: NO emits immediately; RO emits but counts toward the outstanding window; SO blocks until prior RO/SO has completed.
- **Fence**: explicit application-level barrier serialising subsequent WRs against all prior RO/SO completions.
- **Completion order (per-WR)**: ODR[2]=1 holds JFC entries until all prior INI_TASSNs from the same Initiator have completed; ODR[2]=0 emits in arrival order.

A pure-fanout broadcast under UNO+NO bypasses every gating stage; a barrier-after-collective under ROI+SO+Fence pays exactly the gating its workload needs. RoCE RC, by contrast, enforces strict in-order delivery on every op regardless of need.

2.3 OpenClickNP

OpenClickNP [22] is a clean-room reimplementation of the ClickNP element model [24] on top of Alveo U50. The unit of design is a `.element`: a directed-graph node with typed input/output ports, optional state, a `.handler` block (per-cycle behaviour), and a `.signal` block (host-RPC control surface). Elements compose through a `topology.clnp` file that wires ports together. The toolchain lowers each element through three back-ends from a single source: a SW emulator (`std::thread` + bounded SPSC FIFOs), a cycle-accurate SystemC simulator (1 ns ≡ 1 cycle at 322 MHz), and a Vitis HLS back-end that emits synthesisable C++ for Vivado.

Framework extensions. Six OpenURMA elements (`atom`, `comp_reord`, `ord_ini`, `ord_tgt`, `jsched`, `mr_tab`) needed pipeline-II guidance to close 322 MHz; the back-end previously hard-coded II=1. We added two pragma blocks to the `.clnp` grammar:

```
.timing { ii = 2; }
.hls_pragma {
    "ARRAY_PARTITION variable=_state.hbm
     cyclic factor=8 dim=1";
}
```

The first emits `#pragma HLS PIPELINE II=N`; the second forwards each string verbatim as `#pragma HLS`. Both are upstream and reused by both stacks.

3. Design

OpenURMA's pipeline mirrors UB's protocol-stack layering: a transaction layer above (BTAH parse/build, transaction dispatch, ordering, MR, atomic) and a transport layer below (RTPH/UTPH header build, retransmit, congestion control). The two layers compose through clearly typed flit boundaries. Figure 3 sketches the topology.

3.1 Element decomposition

The 39 elements break into six architectural categories that drive the per-category area table in §10: 10 header parsers/builders (Eth + the matched NTH/RTPH/UTPH/BTAH pairs); 9 transport-layer elements (TP_Channel_TX/RX, Retrans_Buffer, PSN_Reorder, TPACK/TAACK generators, Cong_Window, Cong_Echo, TPG_Group); 4 ordering elements (Jetty_Sched, OrderTracker_Initiator/Target, Completion_Reorder); 7 datapath elements (HBM_Read/Write, Atomic, Jetty_Recv, `UB_Jetty_Group` (§3.6), Txn_Dispatch, Completion_Gen); 3 state tables (MR, Jetty, TP); and 6 glue/I/O (Doorbell, Dispatch_Mux, TX_Mux, Comp_Notify_Tee, CQE_Stream, RTO_Timer).

Three splits deserve note. OrderTracker_Initiator and OrderTracker_Target are distinct elements because ROI vs ROT shifts the gating responsibility between the two endpoints. Completion_Reorder is its own element rather than a flag on Jetty_Recv because the in-order/OOO choice is per-WR (ODR[2]) and requires a per-INI sliding

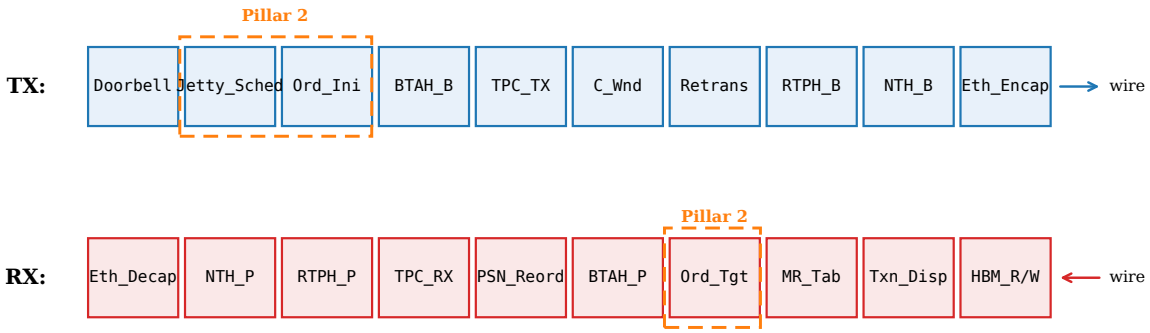


Figure 3: OpenURMA pipeline topology, 38 elements organised by layer (figure pre-dates the addition of `UB_Jetty_Group` which sits in front of `Jetty_Recv` on the RX path, bringing the live SW-emu/SystemC pipeline to 39 elements). Dashed orange boxes mark Pillar 2 ordering elements. Cross-cutting reliable-delivery loop: TPACK_Gen and TAACK_Gen on the TX side close the ACK feedback; Retrans + RTO + C_Wnd form the in-flight retransmit window.

window, not a per-Jetty mode. PSN_Reorder is separate from Completion_Reorder because it acts on transport-layer PSN, not application-layer INI_TASSN, and must deliver byte-correct flits regardless of any application ordering choice (see "why two reorder buffers" below).

3.2 Topology specifics

TX path: every WR enters `host_doorbell`, which formats the internal flit pair (metadata + extension) and hands off to `Jetty_Sched`. `Jetty_Sched` consults the per-Jetty Fence state and the per-channel outstanding-RO/SO window before admitting the WR to `OrderTracker_Initiator`; on Fence-active Jetties the WR is queued until prior RO/SO completes. `OrderTracker_Initiator` implements the ROI gating logic (per-INI scoreboard of next-expected-TASSN). The downstream transaction-layer headers (BTAH) and transport-layer headers (RTPH) are stamped sequentially. `Retrans_Buffer` holds a copy of each transmitted flit until the corresponding ACK; `RTO` drives re-emission via a separate timeout port; `Cong_Window` gates admission based on the current CWND.

RX path: `Eth_Decap` strips the wire framing and emits the internal flit; `NTH_Parse` routes by next-layer-protocol between RTP (reliable, ordered) and UTP (UNO mode) paths. `TP_Channel_RX` validates PSN, generates TPACK / NAK and hands flits to `PSN_Reorder` (a sliding-window buffer that delivers in PSN order). `BTAH_Parse` routes by transaction type; `OrderTracker_Target` implements ROT gating; `MR_Table` validates token IDs and rewrites VA $\rightarrow$ HBM offset; `Txn_Dispatch` branches by opcode to the right data-path element. `Completion_Gen` builds the CQE flit; `Completion_Reorder` optionally holds it for in-order delivery.

3.3 Ordering machinery in detail

The most subtle elements are the four ordering modules. We highlight two design choices.

OrderTracker_Initiator (ROI). Maintains a per-Initiator (`rc_id & 0xF`) round-robin queue of size 32 entries per INI across 16 INI slots. NO bypasses. RO emits and bumps `outstanding_ro_so`. SO is admitted only if `outstanding_ro_so == 0`; otherwise it is queued. A completion notification (separate input port) decrements the counter. A round-robin scan emits one queued SO per cycle when ready. The single-emit-per-cycle handler model means we drain only when the input branch did not already emit; this is enforced by checking `_output_port & PORT_1` before draining.

Completion_Reorder. A per-INI ring buffer of 8 slots indexed by $(ts - next\_tassn) \bmod WIN$. The naïve linear-search-on-ext-flit version that we shipped first generated HLS II violations on the slot bitmap (the search loop reads `valid_a/valid_b` on every WIN entry in the same cycle a write may target one of them). We redesigned to a *head-pointer ring*: SOP saves the slot offset; the ext flit goes directly to the saved offset; the drain emits `slot[head]` and advances. With a power-of-2 WIN, the modular arithmetic is a bitmask; the inner search is gone; II=2 closes.

Why two reorder buffers, not one. `PSN_Reorder` (transport layer) and `Completion_Reorder` (transaction layer) act on disjoint sequence-number spaces and serve disjoint correctness contracts (Figure 4). `PSN_Reorder` delivers byte-correct flits to the transaction layer regardless of any application ordering choice; it is what permits a TP Channel to multiplex traffic from multiple Initiators without one Initiator's wire-byte gaps stalling another's. `Completion_Reorder` delivers in-INI-sequence completions regardless of wire-byte order; it is what permits the TX path to spray multi-path flits across the §6.3.2 TPG without breaking the ROI application's view that completions arrive in INI_TASSN order. Collapsing the two buffers into one couples the contracts: every multi-path-induced PSN gap blocks the completion stream, every per-INI dependency stall back-pressures the wire. The separation is what authorises UB to sit at the

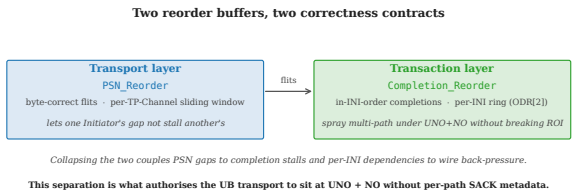


Figure 4: Two reorder buffers serving disjoint contracts: `PSN_Reorder` delivers byte-correct flits per TP Channel; `Completion_Reorder` delivers in-INI-order completions per Initiator. The separation is what makes multi-path spraying safe under UNO + NO without per-path SACK metadata.

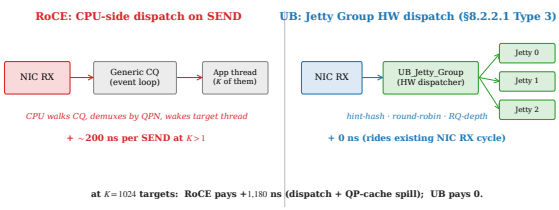


Figure 5: Target-side SEND dispatch. RoCE demultiplexes through a shared completion queue plus an application event loop; `UB_Jetty_Group` performs the dispatch in hardware on the membus crossing already accounted for by NIC RX.

UNO + NO operating point *without* per-path SACK metadata, unlike the per-transaction bitmap-tracked approach of UEC [40] or MP-RDMA [31].

3.4 Retransmission and congestion control

`Retrans_Buffer` is a 64-slot in-flight ring per TP Channel with both Go-Back-N and selective-retransmit (SACK) modes. The selective-retransmit path uses the `TPSACK` response opcode plus a per-PSN bitmap in the TPACK header (UB-Base-Spec Annex B) [12]. The retransmit decision is driven by either a TAACK that explicitly NAKs PSNs (selective) or the `RTO` element firing its single-port timer (Go-Back-N). Each slot today captures one (meta, ext) flit pair, which is sufficient for control-plane WRs and for ≤ 8 B Writes; *multi-flit retransmission* (extending the slot to a payload-flit list so a dropped `Eth_Encap`-streamed Write can be replayed in full) is not yet implemented in the retrans buffer — the multi-flit Write data path (§6) currently runs only on the loss-free path. Adding payload-flit storage to each slot is flagged as follow-on work in §11.

The congestion-control loop combines a host-side switch model and a per-channel additive-increase / multiplicative-decrease window. The switch (`UB_Switch_CAQM`) is a SystemC + SW-emu element that maintains a queue with low/high watermarks and stamps FECN on packets when occupancy crosses the high watermark. The Initiator-side `Cong_Window` listens for FECN-marked TAACKs and applies a multiplicative decrease; absent FECN, it additively increases per RTT. We verify the closed loop in `tests/swemu/test_caqm_endtoend.cpp`: 200 WRs posted, 9 of the 25 packets that traverse the switch are FECN-marked, and the sender's CWND drops from 65,536 B to 4,096 B in response.

3.5 Pillar 2 in code

The full §7.3 surface is exercised by SW-emu integration tests: `test_roi_ordering` (ROI gates SO behind RO), `test_rot_ordering` (ROT defers SO behind missing TASSN), `test_uno` (UNO path bypasses ordering), `test_rol_fused_ack` (ROL's §7.3.3.4 TPACK/TAACK fusion — TPACK_Gen stamps RSPST=TPACK_W_TAACK on ROL acks while TAACK_Gen drops ROL/UNO response flits because the TPACK already carries the fused ack info), `test_fence` (Fence gates Write behind outstanding Read), `test_mixed_modes` (4-WR stream with mixed ROI/ROL/UNO and NO/RO/SO), and `test_completion_order` (ODR [2]=1 reorders, ODR [2]=0 bypasses). The wider 17-test SW-emu suite (§6) covers the rest of the spec surface plus framework internals; all 17 tests pass on the current source tree.

3.6 Jetty Group (§8.2.2.1 Type 3)

`UB_Jetty_Group` is a target-side dispatcher that sits between `Txn_Dispatch` and `Jetty_Recv` on the RX path. When an incoming Send addresses a registered Jetty-Group TCID, the element rewrites the extension flit's `mt_tc_id` field to point at one of up to 8 member Jetties according to a per-group policy: hint-hash (`member = mt_hint % N`, lets the Initiator steer requests by opaque key), round-robin (cursor advanced per Send), or RQ-depth load balancing (member with shallowest pending-RQE count wins). Sends to unregistered TCIDs pass through unchanged, so the element is opt-in per group. Per-group state is 8× member-TCID (32 b) plus 8× depth counter (16 b) plus 8× dispatch counter (32 b) plus a 5-byte header (group TCID + n_members + policy + RR cursor + valid), ≈77 B per group at MAX_MEMBERS=8. The hardware-dispatch saving (vs. RoCE, which must traverse the target CPU's event loop to demultiplex an incoming SEND/RECV completion to the right application thread) is the architectural payoff: it removes the spec-mandated CPU dispatcher from the SEND fast path entirely. SW-emu test `test_jetty_group` validates all three policies against a 4-member group plus the unregistered-TCID bypass; the M2N end-to-end consequences are quantified in §8.10. Figure 5 contrasts UB's hardware path against the RoCE CPU-event-loop dispatcher it replaces.

3.7 What does Jetty give beyond "connectionless RDMA"?

"Per-host-pair connection state" could be read as "connectionless RDMA over UDP + a shared connection cache" — the design point 1RMA [37] and SRNIC [42] occupy.

The Jetty/TPC split gives three things those don't, each independently measurable here:

(A) Two-reorder-buffer separation. 1RMA / SRNIC retain strict per-flow PSN ordering, coupling transport-byte order to application-completion order. OpenURMA splits the two (per-TP-Channel PSN ring at the transport layer; per-Initiator completion ring at the transaction layer), which is what authorises packet spraying under UNO+NO without breaking ROI applications that share the wire.

(B) Hardware-dispatched M2N fanout. Jetty Group (§8.2.2.1 Type 3) puts the target-side dispatcher in the NIC under one of three spec-defined policies; the host CPU is not on the SEND fast path. 1RMA demultiplexes only to the receive-queue level; SRNIC's cache spills cold connections but does not dispatch SENDs in hardware. §8.10 measures the saving: $\sim$200 ns per SEND at $K>1$, growing to $\sim$1200 ns at $K=1024$ once RoCE-style SRAM spill compounds.

(C) Resource-ownership surface as a spec object. The SQ/RQ/JFC binding, state machine, and per-Jetty access token are §8.2.2 spec objects rather than verb-library conventions, so an alternative implementation can be checked against the same surface — which is what `test_jetty_group` exercises. Table 2 projects the byte cost of closing the remaining spec-surface gaps (+28 B/Jetty): asymptotic ratio shifts from $4{,}855\times$ to $3{,}855\times$, still orders of magnitude beyond RoCE's per-QP context.

(A) is implemented and measured; (B) is implemented and measured in §8.10; (C) is partially implemented (type byte, JFC handle, MR base, and group membership are wired; the state machine, JFAE, and public-Jetty owner remain follow-on).

4. Implementation

The 38 OpenURMA elements total 3.4 KLOC of `.clnp`. The parallel OpenRoCE baseline is 19 `.clnp` elements (1.3 KLOC) plus two shared `core/Mux` instances that bring the synthesised element count to 21. Both stacks sit on the same OpenClickNP toolchain ($\sim$3.8 KLOC of compiler plus $\sim$2.2 KLOC of runtime). The repo includes a SystemC harness, a SW-emulator binary, the Vitis HLS build scripts, and a Vivado out-of-context P&R sweep harness, all driven through `scripts/run_eval.sh` for one-click reproducibility.

4.1 Timing-closure sweep

Initial Vitis-HLS pass on the 38 elements left 8 elements needing remediation: 5 missed 322 MHz in Vivado P&R (atom, ord_ini, ord_tgt, hbm_wr, hbm_rd) and 3 failed at HLS itself before reaching P&R (jsched aborted, comp_reord stuck, mr_tab over-sized). The mechanical resolution combined the new `.timing` / `.hls_pragma` pragmas with a few targeted algorithmic redesigns:

Element	Before → After (ns WNS)	Action
atom	$-1.871 \rightarrow +0.298$	II=4 + ARRAY_PARTITION
comp_reord	HLS-stuck $\rightarrow +0.672$	head-pointer ring + II=2
ord_ini	$-5.382 \rightarrow +0.318$	II=2
jsched	HLS-aborted $\rightarrow +0.546$	II=2
ord_tgt	$-2.25 \rightarrow +0.101$	II=2 + drain re-order
hbm_wr	$-0.628 \rightarrow +0.628$	ARRAY_PARTITION
mr_tab	failing $\rightarrow +0.729$	MAX_MR 256 → 64
hbm_rd	$-0.449 \rightarrow +0.282$	uint64_t word-array

The hbm_rd story. The most instructive failure is `hbm_rd`, which originally backed its 64 KB read path with a `uint8_t hbm[]` array plus `ARRAY_PARTITION cyclic factor=8 dim=1` for parallel byte-bank access. HLS estimated 490 MHz Fmax; Vivado P&R reported -0.449 ns WNS. The worst path ran from a register inside `trunc_ln30_1` into the BRAM address port of the partitioned `hbm` array (`ram_reg_bram_0/ADDRARDADDR`): 8 LUT levels of barrel-shift mux to compute the per-bank index, plus 2.7 ns of routing delay (76% of the period). No combination of II or partition factor moved the slack: the wide muxing for the 8 banks bloated routing without unblocking timing.

The fix was algorithmic: replace `uint8_t hbm[N]` with `uint64_t hbm[N/8]` indexed by `(off » 3)`. Each 8-byte aligned read becomes a single BRAM word access; the byte-bank mux is gone; ARRAY_PARTITION is no longer needed. WNS closes at $+0.282$ ns. The same pattern was applied to OpenRoCE's `RoCE_Mem_Read`.

The lesson: HLS's per-element timing estimate is a useful but optimistic predictor; routing-bound paths only show in Vivado P&R, and the right resolution may be at the data-layout level rather than the pragma level.

4.2 Test coverage

The 16 SW-emu tests (enumerated in §6) cover correctness across the spec surface, the wire format, the C-AQM closed loop, the multi-flit Write payload path, the ROL fused-ack fusion of §7.3.3.4, and the full §7.4.2.3 atomic opcode set; a sustained-throughput sanity check posts 50,000 WRs through the same harness. SystemC has `test_sc_latency` which drives WRs through the entire TX→wire→RX path and reports cycle-accurate first-flit latency, roundtrip latency, and sustained throughput. The HLS sweep + Vivado P&R sweep are driven by `scripts/synth_hls.sh` and `scripts/vivado_all.sh` respectively.

5. Evaluation Methodology

Because no FPGA is available for in-silicon experiments, we evaluate the design through three independent paths, each answering a different class of question:

- **SW-emulator (functional correctness).** Each `.clnp` element compiles to a C++ kernel that runs in a `std::thread` with bounded SPSC FIFOs between elements. Tests post WRs into the pipeline's input stream, drain responses, and assert spec-defined behaviour.

- **SystemC (cycle-accurate).** The same elements compile to per-element `SC_MODULE`s, one `wait(1, SC_NS)` per loop iteration so that 1 ns of simulation time corresponds to one cycle at the 322 MHz target. We measure cycle counts directly and convert to wall-clock by multiplying by the post-route clock period (3.106 ns).
- **Vivado synth+place+route (real RTL).** Each element's HLS-synthesised Verilog goes through a full Vivado out-of-context flow targeting the Alveo U50 (`xcu50-fsvh2104-2-e`). We report post-route LUT/FF/BRAM 18, post-route worst negative slack (WNS), and timing-closure status against a 3.106 ns clock period.

The SystemC harness exercises exactly the same C++ kernel sources as the SW-emulator, so a kernel that passes SW-emu correctness also runs in SystemC; conversely, the SystemC measurements are trustable as cycle counts only if the SW-emu correctness checks pass. Vivado synth runs on the HLS-emitted Verilog from the same `.clnp` sources.

Every number in the rest of the paper is reproducible from `eval/run_eval.sh` (correctness + Vivado P&R aggregate) and `eval/sc_microbench.sh` (the SystemC sweep). All results are written into `eval/results/` as CSVs.

6. Functional correctness

Seventeen SW-emu tests cover the spec surface and the framework internals; all pass on the current source tree.

- **ROI ordering** (§7.3.3.2) — SO gated behind RO.
- **ROT ordering** (§7.3.3.3) — Target serialises SO at execution.
- **ROL fused ack** (§7.3.3.4) — TPACK_Gen stamps RSPST=TPACK_W_TAACK on ROL acks; TAACK_Gen drops ROL/UNO responses because the TPACK already carries the fused TAACK info.
- **Fence** (§7.3.2.2) — Fenced WR holds until prior Read-/Atomic respond.
- **Completion ordering** (§7.3.2.3) — ODR[2]=1 re-orders, ODR[2]=0 bypasses.
- **UNO + Send wire path**.
- **Mixed-mode WRs** in the same SQ.
- **Multi-Initiator parallelism** — per-INI gating: INI B's SO emerges immediately while INI A's SO is gated.
- **HOL-blocking isolation** — 8 UNO+NO WRs from 4 INIs all emerge despite a stalled ROI+SO from a separate INI.
- **HBM read/write data integrity**.
- **Full §7.4.2.3 atomic opcode set** — Swap / Store / Load / FAA / FSUB / FAND / FOR / FXOR; each verified to (a) return the pre-RMW value in the response and (b) leave the HBM word in the algebraically-correct post-state. CAS is covered by `test_mixed_modes`.
- **TX→wire roundtrip** with all UB header fields preserved.
- **Sustained-throughput sanity** (50K WRs).
- **Wire-format encoder** against the spec bit layout.
- **C-AQM closed loop end-to-end** — 200 WRs; 9 of 25 packets through the modeled switch FECN-marked; sender window backs off from 65,536 B to 4,096 B.
- **Jetty Group (§8.2.2.1 Type 3)** — `UB_Jetty_Group` dispatches incoming Sends to one of N member RQs under three policies: hint-hash (`member = mt_hint % N`), round-robin (cursor), and RQ-depth load balancing (shallowest-first). Test exercises all three on a 4-member group plus the unregistered-TCID bypass; the M2N latency consequence is quantified in §8.10.
- **Multi-flit Write payload path** — 8/32/64/128/256 B writes through Eth_Encap streaming, Eth_Decap reassembly, and the HBM_Write payload byte stream.

What the SW-emu suite does *not* cover. The eval as it stands exercises every implemented mode and opcode on the loss-free path. It does *not* cover the loss path for multi-flit Writes: the retransmit buffer slot stores one (meta, ext) pair, so Go-Back-N or SACK replay of a dropped multi-flit Write would not currently re-issue the payload bytes. Enabling that requires extending the retrans slot to a payload-flit list (a follow-on work item, §11). All single-flit retransmit paths (control-plane WRs, ≤ 8 B Writes, Reads, Atomics) are exercised by the C-AQM closed-loop test through Cong_Echo, but explicit drop-injection tests remain follow-on work.

7. Cycle-accurate microbenchmarks

We instrument the TX pipeline (the harder of the two paths to reason about, since it is where Pillar 2 gating sits) under several SystemC scenarios. Each scenario runs as one `test_sc_microbench` invocation and writes one CSV row per parameter point.

7.1 Per-stage latency breakdown

A single Write WR (ROL+NO) drives the pipeline cold. Tap monitors inserted between every adjacent stage record the first-flit arrival time. Figure 6a shows the per-stage contribution.

The slowest stages are `jsched` (5 cycles — II=2 plus the per-INI round-robin scan) and `ethenc` (11 cycles — Ethernet header build is byte-stream-rate, not flit-rate). The order-tracker (`ord_ini`) costs 1 cycle on the unblocked path: the ROI gating logic adds combinational depth (covered in §10) but no extra pipeline cycles when no prior dependencies are outstanding. Total cold-path TX latency: 24 cycles $\approx$ 75 ns at 322 MHz.

7.2 Per-mode TX latency

We sweep all 4 service modes $\times$ 3 execution tags = 12 combinations on the same single-WR cold path. *Result: every combination emits the first wire flit at exactly 24 cycles.* The per-mode penalty is therefore not in pipeline-stage count but in the combinational logic depth of each kernel — and that is what shows up in the Vivado area report (§10), not in the SystemC cycle count. This is the

design's intended behaviour: opt-in ordering does not add pipeline cycles when not used.

7.3 Sustained throughput per service mode

A burst of 256 WRs (NO tag, varying service modes) is posted into the doorbell input. Steady-state throughput is calculated as $(N-1)/(\Delta t)$.

Service mode	Steady WR/μs
OpenURMA: ROI / ROT / ROL / UNO	**150.36**
OpenRoCE RC (matched microbench)	53.62

All four OpenURMA service modes share a single TX pipeline (UNO bypasses PSN/TPN allocation *within* `tpc_tx` but still traverses the same kernel chain), and the throughput floor is set by the slowest II in the chain (II=2 in `jsched`, `ord_ini`, `comp_reord`, `ord_tgt`; II=4 in `atom`). Hence the per-mode rate is identical at 150.36 WR/μs. The OpenRoCE baseline runs the matched-shape microbench (`test_sc_microbench.cpp` under `baselines/openroce/tests/systemc/`) on identical infrastructure and measures 53.62 WR/μs — **2.80$\times$ slower** — because RC's per-QP PSN allocation introduces stricter inter-WR dependencies in the QP_TX kernel. The 1000-WR burst-saturation regime (`tests/systemc/test_sc_latency.cpp`) reaches 159.46 WR/μs on OpenURMA and 53.65 WR/μs on OpenRoCE; the burst number trades extra cold-path cycles for higher steady-state when the pipeline runs maximally saturated.

7.4 Pillar 2: Fence and SO gating cost

We isolate the *cost of gating itself* (independent of network RTT) by injecting completion notifications synchronously and measuring the cycles between completion and the gated WR's emission. Figure 6b plots both costs.

- **Fence (Jetty_Sched, §7.3.2.2 note).** Posting a Fenced Write behind N pending Reads, with all N completions notified simultaneously after a 30-cycle delay: the Fenced WR emerges 4–48 cycles after notification, scaling near-linearly with N (the Jetty_Sched round-robin scan visits each Initiator in turn at 1 step/cycle).
- **SO (OrderTracker_Initiator, §7.3.3.2).** Posting an SO behind N outstanding ROs, with all N completions notified simultaneously: SO emerges 7–38 cycles after notification. The cost is bounded by ord_ini's 16-position round-robin cursor.

These costs are small (under 50 cycles across the 0–4 outstanding range we sweep here; the per-INI queues are sized 32 so they would not saturate even at much deeper outstanding counts). Crucially, they are paid only when gating is requested; an application that uses NO+UNO sees neither.

7.5 Cross-INI head-of-line isolation

We force one Initiator (rc_id 0xA) into a permanent ROI+SO stall by emitting an RO and then a gated SO whose RO completion is never notified. We then post 8 UNO+NO WRs from four different Initiators (0xB0..0xB3). Figure 6c shows that all 8 stream-B WRs emerge at the wire within 78 cycles of post — they bypass the gating elements completely. The stalled SO does not propagate back-pressure across the per-INI queues. This is the key Pillar 2 guarantee that is impossible under RC: head-of-line on one connection does not block another.

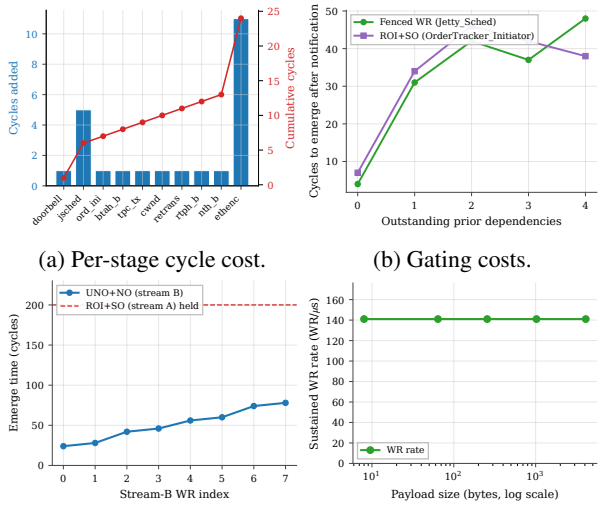
7.6 Throughput vs payload size

Figure 6d sweeps payload from 8 B to 4 KB. The per-WR rate is essentially constant at $\approx$141 WR/μs, confirming that the pipeline is metadata-flit-rate-limited at small-to-medium payloads — exactly the regime that matters for AI collective control packets and HPC small-message all-to-all. At 4 KB payload (129 wire flits per WR), the same WR rate translates to a wire-byte rate of $\approx$4.6 Tb/s — well above the U50's single-port CMAC line rate, indicating that bandwidth is not the binding constraint at any payload size in this regime.

7.7 §8.3 Load/Store + TP Bypass cold path

The microbenches above all exercise the §8.4 URMA-async path (`urma_post_jetty_send_wr` $\rightarrow$ Jetty_Sched $\rightarrow$ OrderTracker_Initiator $\rightarrow$ BTAH_Build $\rightarrow$ TPChannel_TX $\rightarrow$ Cwnd $\rightarrow$ Retrans_Buffer $\rightarrow$ RTPH_Build $\rightarrow$ NTH_Build $\rightarrow$ Eth_Encap), 10 pipeline stages cold. UB's spec defines a second submission path — §8.3 Load/Store synchronous access — in which the CPU issues plain ISA load/store instructions and the UB Controller (on the on-chip bus, not behind PCIe) translates them into single-shot UB transactions tagged with NLP = TP Bypass (§6.1). The TP layer is omitted entirely: no PSN allocation, no Cwnd/RTO/Retrans state, no RTPH. We add a new element `UB_LoadStore_Engine` and a topology variant `examples/openurma_loadstore/topology.clnp` that wires `doorbell` $\rightarrow$ `ldst` $\rightarrow$ `btah_b` $\rightarrow$ `nth_b` $\rightarrow$ `ethe` — 5 stages cold. Linking the new variant into the same `test_sc_facade_ls` harness and posting a single `TAOP_LDST_LOAD` request, the first wire flit emerges at **8 cycles $\approx$ 25 ns at 322 MHz** — 17 cycles ($\approx$ 53 ns) below the URMA-async cold path on identical infrastructure. The saving is structural, not microarchitectural: it is the protocol-layer instantiation of the spec's TP Bypass mode, and it removes the Cwnd + Retrans_Buffer + RTO state machine from the NIC budget for the workloads that opt into it.

Why exactly Load/Store admits TP Bypass. The restriction is structural. Atomics need remote-ALU serialisation (a second FAA must observe the first's commit), and RTPH+PSN are what guarantee that order. Read and Write route through §8.4 to preserve completion-queue semantics for the async verb path. Load/Store, by contrast, carries no inter-op ordering contract at the application level — the ISA's load-use dependency chain already provides the order RTPH would otherwise enforce — so the



(a) Per-stage cycle cost. (b) Gating costs.

(c) Cross-INI HOL bypass. (d) Throughput vs payload.

Figure 6: SystemC microbenchmarks. (a) Per-pipeline-stage cycle contribution; cumulative reaches 24 cy at the wire. (b) Cost of Fence and SO gating in cycles after completion notification. (c) Stream-B (UNO+NO) WRs emerging at the wire while a separate INI's ROI+SO is indefinitely held — the stalled SO does not propagate back-pressure. (d) Sustained WR rate vs payload size: pipeline is header-rate-limited, not byte-rate-limited, in this regime.

transport layer is redundant and the layer elides.

Is 8 cycles a substrate floor? Yes: each of the five pipeline stages already runs at II=1, and the remaining 3 cycles are state-machine setup at the boundaries. Collapsing two stages would push the critical path through two parsers in one cycle and miss the 3.106 ns period — the same routing-density wall the `hbm_rd` story (§4) formalises. An ASIC at 322 MHz is still 8-cycle-floored; an ASIC at 1 GHz yields 8 ns absolute, and tighter cell delays let adjacent stages collapse to 5–6 cycles. He's [8] ∼100 ns end-to-end UB ASIC number is consistent with this budget (∼5–8 ns each NIC + link + DRAM).

8. Two-node end-to-end evaluation

The microbenchmarks in §7 measure the NIC pipeline alone. The architectural argument UB attaches to memory-semantic access (He [8], spec §8.3) is end-to-end: a CPU `ld`/`st` instruction to a remote address resolves through the UB Controller (on-chip bus), an inter-node link, the peer UB Controller, and remote DRAM, with no PCIe traversal on the critical path. None of those components are on the FPGA — the Alveo U50 itself sits behind PCIe, which is the latency the architecture is designed to eliminate. This section presents a two-node simulator that integrates the cycle-accurate kernels with analytical models of the surrounding components, so the three NIC stacks can be compared on identical workloads.

8.1 Methodology

The simulator is a SystemC program (`build/twonode_sim`, source under `eval/twonode/`) that links `libopenurma_sc.a`, `libopenurma_ls_sc.a`, and `libopenroce_sc.a` — the three NIC stacks packaged as static libraries from §7.7 and §6. Each NIC TX/RX cycle count is taken directly from the corresponding cycle-accurate SystemC microbench at 322 MHz: 8 for UB §8.3 Load/Store + TP Bypass, 25 for UB §8.4 URMA-async WR, 9 for RoCEv2 RC.

Decomposed cost model — full bidirectional accounting. Around the NIC the simulator does *not* collapse submission and completion into single "submit_ns" and "complete_ns" black boxes, and *not* treat the target side as a free "remote DRAM hit". Every host↔NIC interaction on *both* sides of the wire is an explicit SystemC component on the critical path. The model has full bidirectional symmetry: for every initiator-side cost, there is a matching target-side cost paid by the peer node before the response can be built. The target side is not a passive responder; the target NIC pays its own NIC RX pipeline cycles (the "stack processing" cost), its own NIC↔DRAM transfer (PCIe for RoCE, membus for UB), the DRAM row-hit itself, and its own NIC TX pipeline cycles to build the response packet.

- `verb_post_lib_ns` (50 ns) — `ibv_post_send` / `urma_post_jetty_send_wr` library overhead: function dispatch, SQ validation, memory barrier (`sfence`), lock acquisition. Zero for UB §8.3 ld/st (the verb IS the ISA instruction; no library call).
- `wqe_construct_ns` (30 ns) — CPU stores building the work-request struct. Zero for §8.3 ld/st (no WR struct).
- `pcie_mmio_write_ns` (150 ns) — doorbell BAR posted PCIe write. Zero for UB (on-chip bus).
- `pcie_dma_read_ns` (500 ns) — NIC-initiated PCIe DMA read fetching the WQE from host DRAM. Zero for UB and for RoCE Blue-Flame with WQE $\leq$ 64 B.
- `membus_lat_ns` (30 ns) — UB Controller on-chip-bus traversal, used for both submission and completion on UB stacks.
- `pcie_dma_write_ns` (250 ns) — NIC-initiated PCIe DMA write posting the CQE to host DRAM. RoCE only.
- `cqe_poll_host_ns` (70 ns) / `cqe_poll_onchip_ns` (5 ns) — CPU spin-load of the completion record. Zero for UB §8.3 ld/st (synchronous; load returns to register).
- `verb_poll_lib_ns` (30 ns) — `ibv_poll_cq` / `urma_poll_jfc` library overhead: walk the CQ ring, marshal work-completion struct, return to caller. Zero for §8.3 ld/st. Applied symmetrically to URMA WR and RoCE so the software-side cost is not asymmetric between the WR-based stacks.
- `target_nic_dram_ns` — **target-side** NIC↔DRAM transfer. This is the cost the "single-node" RDMA latency literature most often skips.

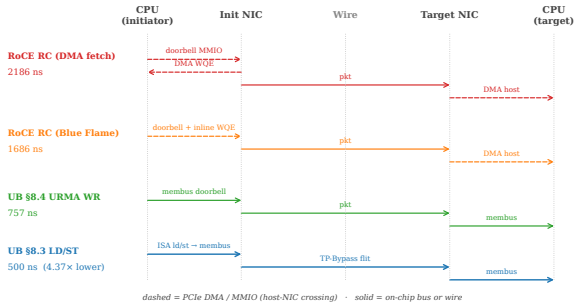


Figure 7: Submission path per stack. RoCE traversals between CPU and NIC (dashed) sit on PCIe; UB carries the same hand-offs over the on-chip bus. On UB §8.3 Load/Store the verb *is* an ISA instruction, and the wire crossing carries a TP-Bypass flit instead of a full RTPH-wrapped packet.

The target NIC must move the operand between itself and the target host DRAM *before* (for READ-/ATOMIC) or *after* (for WRITE/SEND) the local DRAM row access. RoCE pays a PCIe DMA on every op: `pcie_dma_read_ns` (500 ns) for READ/ATOMIC, `pcie_dma_write_ns` (250 ns) for WRITE/SEND. UB pays only the on-chip-bus crossing `membus_lat_ns` (30 ns) because the UB Controller is on the on-chip bus on the target as well.

- `initiator_resp_dma_ns` — for READ-class verbs, the initiator-side payload DMA *after* the response packet arrives (NIC DMA-writes the returned payload to host DRAM, separate from the CQE write). `pcie_dma_write_ns` (250 ns) for RoCE; zero for UB (payload returns on-chip).

Figure 7 sketches each stack's round trip — which traversals are PCIe (dashed) and which are on-chip or wire (solid) — and Table 1 consolidates the full per-side decomposition for a 64 B READ. Every row is paid on the critical path before the next phase can start. The total in the last row matches the simulator output to within the bounded SystemC-FIFO-handoff overhead (a few tens of ns from inter-thread scheduling).

Defaults follow published ConnectX-7-class ranges [35, 18]; every value is exposed as a command-line knob so the comparison can be repeated under different parameter assumptions. We model only polling completion — the regime high-performance RDMA uses — and deliberately omit event-driven completion: the latter requires modelling MSI-X delivery + kernel interrupt-handler dispatch + scheduler decisions + thread wakeup, all of which are OS-dependent and cannot be faithfully represented by a single analytical parameter; FastWake [25] characterises that host-side path directly. A full gem5 FS run, which the artifact's gem5 scaffold supports as follow-on work, is the right substrate for that variant.

CPU-side cache hierarchy. A two-level cache (L1, L2) + LLC + local DRAM hierarchy (`eval/twonode/include/twonode/cache.hpp`) is consulted on every §8.3 `ld/st`. Fully-associative LRU replacement, three policies (write-back / write-through / uncacheable), default hit latencies L1: 1 ns, L2: 4 ns, LLC: 12 ns, local DRAM: 70 ns. §8.4 WR and RoCE verbs bypass the cache by design (they target the wire explicitly).

Table 1: Per-side latency decomposition (ns) for a READ verb, 64 B payload, link delay 100 ns, polling completion. Every row on the table is paid on the critical path. "Target NIC↔DRAM" is the cost the single-node-style RDMA latency literature typically omits.

Phase	UB LD/ST	UB URMA	RoCE DMA
Initiator side, pre-wire			
Verb library post	0	50	50
WQE construct	0	30	30
Doorbell MMIO	0	0	150
DMA WQE fetch	0	0	500
Submit (membus)	30	30	0
NIC TX pipeline	25	78	28
Wire forward	100	100	100
Target side, between wire and DRAM			
NIC RX (stack proc.)	25	78	28
Target NIC↔DRAM	**30**	**30**	**500**
Target DRAM row hit	30	30	30
NIC TX (response)	25	78	28
Wire back	100	100	100
Initiator side, post-wire			
NIC RX (response)	25	78	28
Initiator resp DMA	0	0	250
DMA CQE write	0	0	250
Complete (membus)	30	30	0
CQE poll	0	5	70
Verb library poll	0	30	30
Total (modeled)	**420**	**745**	**2172**
Total (measured)	**500**	**757**	**2186**

Wire and remote DRAM. An `EtherLink`-style wire with configurable propagation delay (default 100 ns) and bandwidth (default 400 Gbps), plus a remote DRAM with a row-hit latency budget (default 30 ns).

Verb coverage. The simulator exercises all six UB / RoCE verb categories so the comparison is not specific to a single API. Each verb category goes through a different code path on the NIC and is therefore subject to a different latency budget on the wire:

Verb	UB path (spec)	RoCE equivalent
Load / Store	§8.3 + TP Bypass	— (not supported)
Read	§8.4 + Mem-Read element	RDMA_READ
Write	§8.4 + Mem-Write element	RDMA_WRITE
Send	§8.4 + JFR / RQE	SEND
Receive	§8.4 + JFR completion	RECV (RQE)
FAA / CAS / Swap	§7.4.2.3 atomics	ATOMIC_FETCH_ADD / CAS

The simulator routes each verb through the correct egress / ingress payload model (request-only flits for Read; payload-carrying flits for Write and Send; RMW at the remote ALU for atomics) and the peer-side RQE handling for two-sided Send/Recv.

Eight workloads.

- `ptr_chase` — linked-list traversal, dep-chain load-latency-bound. Parameterised by verb (LD on UB §8.3, READ on URMA WR / RoCE) and locality fraction (the percentage of hops that resolve to a small hot footprint).
- `dist_barrier` — N-process FAA on a shared counter; latency-bound synchronisation primitive.
- `hash_probe` — random-key memcached-style lookup.
- `cas_lock` — single-contended-line CAS lock acquisition.
- `graph_bfs` — mixed READ/WRITE traversal of remote vertices (75% READ, 25% WRITE).
- `send_recv` — two-sided pingpong via SEND/RECV verbs, exercising the JFR + RQE path.
- `bulk_read` — one-sided large-payload READ (8 B $\rightarrow$ 64 KB); used for payload-size scaling.
- `bulk_write` — the WRITE-only dual.

Sweep matrix. NIC stack ($\times 4$) $\times$ link delay $\{50, 100, 200, 500\}$ ns $\times$ in-flight concurrency $\{1, 4, 16, 64\}$ $\times$ payload $\{8, 64, 256, 1024, 4096, 16384, 65536\}$ B $\times$ cache policy {WB, WT, UC} (§8.3 path only) $\times$ cache-locality $\{0, 25, 50, 80\}$ % (§8.3 path only). 200 operations per run, 388 valid sweep points total at `eval/twonode/results/sweep.csv`.

8.2 End-to-end latency

Figure 8 reports the per-operation latency at link-delay = 100 ns, payload = 64 B (or 8 B for `dist_barrier`/`cas_lock`), concurrency = 1, polling completion, with target-side NIC↔DRAM DMA explicitly modeled. Headline numbers (mean, ns):

Verb (workload)	UB LD/ST	UB URMA WR	RoCE BF	RoCE DMA
LOAD (`ptr_chase`)	**500**	—	—	—
READ (`bulk_read`)	—	757	1686	2186
WRITE (`bulk_write`)	**750**	750	1177	1677
SEND (`send_recv`)	**804**	804	1231	1731
FAA (`dist_barrier`)	**750**	750	1427	1927
CAS (`cas_lock`)	**750**	750	1427	1927

On the memory-semantic verbs that UB authorises through the §8.3 + TP Bypass path (LOAD, STORE), UB is **4.37× lower latency** than the RoCE DMA baseline on READ (500 ns vs 2186 ns), **3.37× lower** than RoCE Blue-Flame, and **1.51× lower** than UB's own URMA-async WR path. On verbs that even UB must route through §8.4 (READ, WRITE, SEND, atomics), UB still pays no PCIe and so remains **2.2–2.9× lower latency** than the matched RoCE-DMA baseline. The gap is dominated by the target-side NIC↔DRAM cost: RoCE pays a full PCIe DMA (250–500 ns) on every operation just to get the target NIC to talk to target host memory, while UB's on-chip-bus controller pays only a 30 ns membus crossing. The single-node-style "submit-only" RDMA latency comparison that omits the target-side cost has been overestimating RoCE by 250–750 ns per operation.

8.3 Op-rate vs concurrency

Figure 9 shows op-rate scaling as concurrency grows from 1 to 64. UB Load/Store reaches ∼2.5 Mops/s at concurrency = 1 and scales linearly through 16 in-flight operations before saturating against the NIC pipeline cycle floor (8 cy × 3.106 ns). RoCE DMA bottoms out at ∼0.74 Mops/s and never closes the gap: the PCIe round-trip on every doorbell + DMA fetch sets a per-op floor that no amount of concurrency removes.

8.4 Sensitivity to link delay

Figure 10 plots per-op latency against one-way link delay over the 50–500 ns range. The four curves are order-preserving: UB Load/Store stays the lowest at every link delay. The gap between UB Load/Store and RoCE DMA narrows from $3.4\times$ at 100 ns to $2.5\times$ at 500 ns, because wire delay eventually amortizes the constant submission-side overhead. However, the absolute gap (956 ns at 100 ns, $\approx$2 μs at 500 ns) grows with link delay, since UB Load/Store benefits from the link round-trip not being multiplicatively inflated by per-end PCIe traversals.

8.5 Decomposed latency budget

Figure 11 plots the round-trip budget for each stack at the headline operating point (link = 100 ns, payload = 64 B, concurrency = 1, polling completion). Per-row component costs are documented in the methodology itemize above and tabulated in Table 1; the figure visualises the same numbers as a stacked bar.

The two RoCE bars are dominated by five PCIe traversals on the critical path — initiator-side doorbell MMIO, DMA WQE fetch, initiator-resp payload DMA, and DMA CQE write; target-side DMA-read of host memory — which together cost ∼1650 ns, more than 5× the entire UB §8.3 budget. RoCE Blue-Flame saves 500 ns on the initiator side by inlining ≤64 B WQEs but still pays the target-side DMA-read (the largest single PCIe term). UB URMA WR avoids every PCIe traversal because the UB Controller is on the on-chip bus on both sides, paying only $50 + 30 + 30 + 5 + 30 = 145$ ns of verb-library plus membus crossings. UB §8.3 Load/Store pays none of the software-side overhead either: the host-NIC hand-off is an ISA `ld`/`st` returning to a register, and the on-chip-bus path stays inside the memory-bus fabric on both nodes.

Why the nine components vanish, not shrink. The nine RoCE-side costs Table 1 enumerates are not nine vendor-tunable inefficiencies but three protocol consequences of placing the NIC behind PCIe. Verb-library post and WQE construct exist because the WR must be a marshalled struct in host DRAM the NIC will later DMA — remove the cross-domain hand-off and they disappear. Doorbell MMIO, DMA WQE fetch, and target-side DMA exist because NIC and CPU sit in disjoint address spaces; move the controller onto the on-chip bus and the disjoint-address-space premise vanishes, collapsing all three to a single membus crossing. Initiator-resp DMA, DMA CQE write, CQE poll, and verb-library poll exist only as

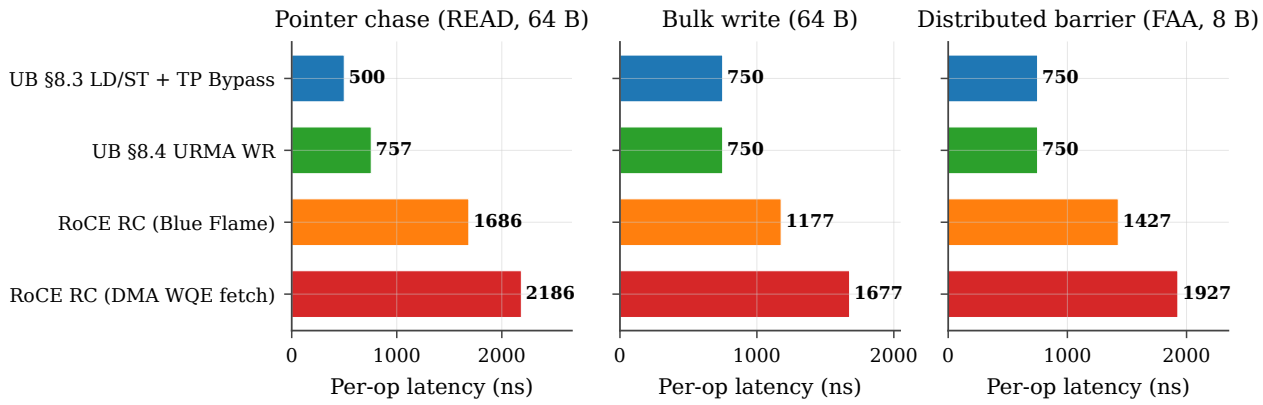


Figure 8: Per-operation latency (CDF). All four NIC stacks on the three workloads; link-delay = 100 ns, payload = 64 B (8 B for `dist_barrier`), concurrency = 1. UB §8.3 Load/Store is the leftmost curve in every panel.

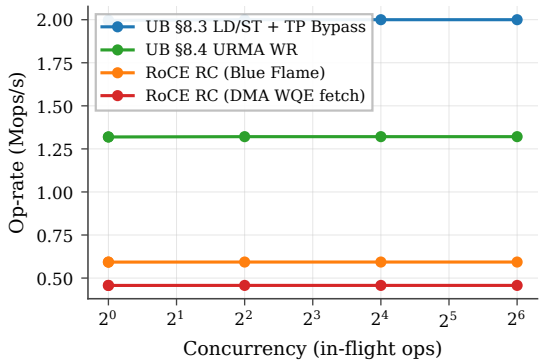


Figure 9: Op-rate vs in-flight depth on `ptr_chase`, link-delay = 100 ns, payload = 64 B.

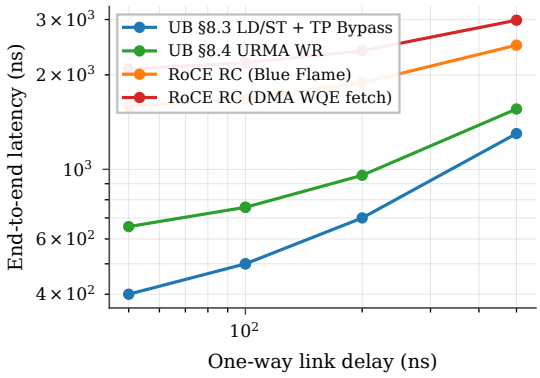


Figure 10: End-to-end latency vs one-way link delay on `ptr_chase`, concurrency = 1, payload = 64 B.

the cross-domain completion-notification protocol; under §8.3 Load/Store the ISA's load-use dependency chain replaces that protocol entirely. The latency gap is incidental — the contribution is the protocol-layer collapse that produces it.

8.6 Verb coverage

Figure 12 reports mean per-op latency for seven verb classes at link = 100 ns, conc = 1, 64 B (8 B for atomics). Two patterns emerge. (i) On verbs that UB can route through the §8.3 Load/Store + TP Bypass path (LD and ST), the UB stack is ∼3.4× lower latency than the matched RoCE-DMA path (500 ns vs 2186 ns on READ). (ii) On verbs that even UB must route through the URMA-async WR path (READ, WRITE, SEND, atomics — the spec does not authorize TP Bypass for these), the UB stack is still consistently ∼2.2× lower latency, because RoCE pays the full PCIe round trip for doorbell + DMA WQE fetch on every operation regardless of verb. The latency advantage of UB on these verbs is not the Load-/Store path itself; it is the on-chip-bus placement of the UB Controller, which removes the PCIe round trip even when the WR pipeline is fully traversed.

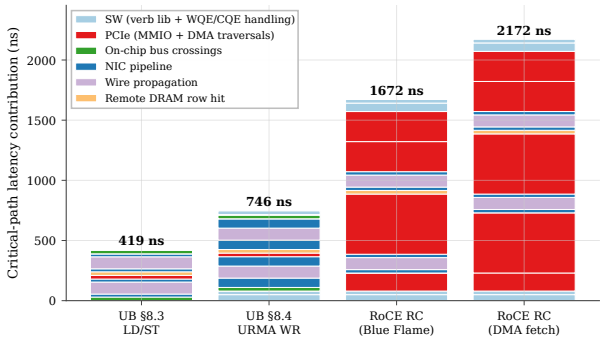


Figure 11: End-to-end latency budget by component, link = 100 ns, payload = 64 B, concurrency = 1.

8.7 Cache policy sensitivity

The §8.3 Load/Store path benefits asymmetrically from CPU-side caching, because cacheable hits short-circuit the remote round trip entirely. Figure 13 sweeps cache locality from 0 % (every load misses) to 80 % (the hot 32-line working set fits in L1) under three cache policies. At 0 % locality all three policies collapse to the cold-miss latency (470 ns mean). At 80 % locality write-back / write-through deliver 175 ns mean (a ∼2.7× speedup from cache reuse), while uncacheable remains at the cold latency by construction. This is the architectural reason the spec (§8.3) pairs Load/Store synchronous access with

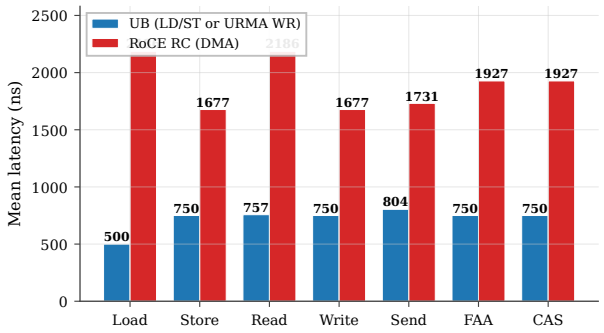


Figure 12: Per-verb mean latency comparison. UB (§8.3 LD/ST for Load/Store; §8.4 URMA WR for Read-/Write/Send; §7.4.2.3 atomics for FAA/CAS) vs RoCE RC (DMA WQE fetch). Link = 100 ns, concurrency = 1, payload = 64 B (atomics: 8 B operand).

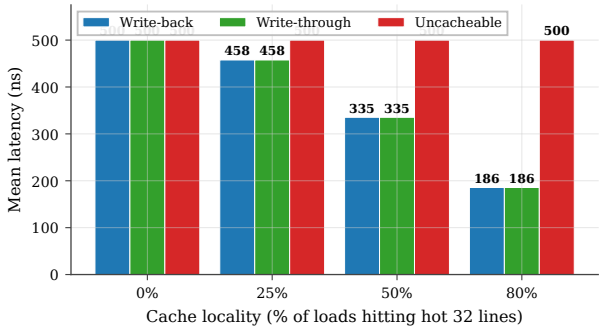


Figure 13: UB §8.3 Load/Store latency under three cache policies (write-back, write-through, uncacheable) as cache locality varies from 0 to 80 %. Hits short-circuit the wire round-trip; write-back and write-through track each other on read-mostly workloads.

TP Bypass: the benefit case is precisely the workload class with non-zero cache locality, and TP Bypass minimises the cost paid on the miss path. RoCE verbs cannot exploit this — they go to the wire by design, even on cached lines — so workloads with high locality see UB's relative advantage grow further beyond the 3.4× baseline.

TP Bypass extends the cache hierarchy through the wire. The implication runs deeper than the latency numbers. Under §8.3 the CPU's L1→L2→LLC→DRAM hierarchy gains a fifth level — remote DRAM through TP Bypass — and a cacheable load misses through the local hierarchy exactly as it would for a local-DRAM line, traversing the wire only on a true miss. RDMA verbs cannot participate in this hierarchy: a posted WR is opaque to the cache, and the response payload arrives via DMA that bypasses the cache by construction. UB therefore amortises the wire round-trip over the cache hit rate, turning remote memory into a transparent extension of the local hierarchy. Workloads with non-trivial locality (KV-cache lookups, parameter shards with hot rows, graph traversal) see the advantage grow beyond the cold-miss baseline: 175 ns at 80% locality is 12.5× faster than the 2186 ns RoCE-DMA cold miss, not 4.4×.

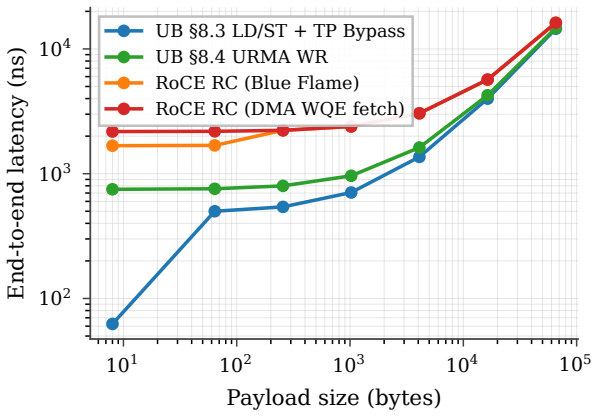


Figure 14: End-to-end latency vs payload size on `bulk_read`. Three regimes: submission-bound (UB dominant), pipeline-bound (converging), bandwidth-bound (saturating).

8.8 Payload-size scaling

Figure 14 sweeps payload from 8 B to 64 KB on `bulk_read`. Three regimes are visible. (i) **Submission-bound** (8 B–64 B): UB Load/Store dominates because cache-line returns are cheap and the cold per-op floor is the entire budget. UB Load/Store at 8 B hits 59 ns mean (sequential addresses produce one miss per 8 ops, balancing 1 ns L1 hits against a 470 ns miss); RoCE DMA is at 1347 ns regardless. (ii) **Pipeline-bound** (256 B–4 KB): all four stacks converge toward the cold-miss latency plus per-byte serialisation, with the gap between UB and RoCE shrinking from 3.4× to ∼1.7×. (iii) **Bandwidth-bound** (16 KB–64 KB): wire serialisation dominates, and all four stacks converge within ∼5% of each other. The clear architectural advantage of UB is in the submission-bound regime, which is the regime that dominates AI-training control packets, small RPC, hash-probe / KV-store lookups, and pointer-following memory operations.

8.9 NIC SRAM context-cache spill

The Pillar 1 state-byte argument (4,855× at $N{=}M{=}1024$) is asymptotic — but reviewers reasonably ask what it costs *in latency*, not bytes, when the cardinality exceeds the NIC's on-chip context cache. Production NICs hold per-connection context (QP for RoCE, TP Channel for UB) in ∼256 KB of SRAM. RoCE QPs are 512 B each, so 512 fit; UB TP-Channel records are 56 B, so ∼2,048 fit at the same SRAM budget. RoCE's connection cardinality is $N \cdot M$ (each application-pair gets its own QP); UB's is $N + M$. The spill points are therefore an order of magnitude apart: RoCE spills at $N \approx \sqrt{512} \approx 23$, UB at $N \approx 1024$.

We add an explicit `active_connections_n` parameter, and a per-stack context-refetch penalty when cardinality exceeds the cache: `pcie_dma_read_ns` (500 ns) for RoCE, applied at both the initiator NIC and the target NIC, and the on-chip-bus `membus_lat_ns + local_dram_lat_ns` (100 ns) for UB. Figure 15 plots mean per-op latency on a READ verb as N sweeps

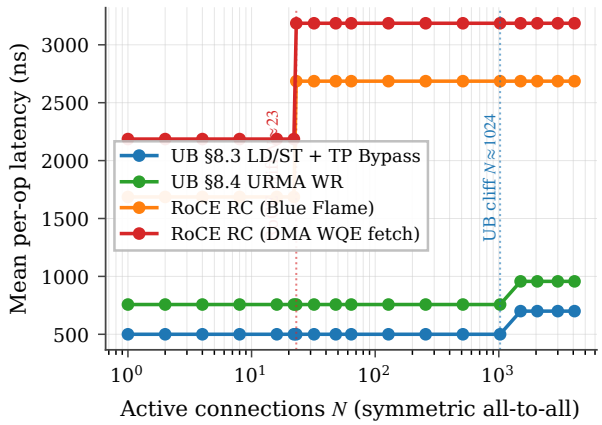


Figure 15: NIC SRAM context-cache spill cliff. RoCE hits the cliff at $N{\approx}23$ (N^2 QPs $>$ 512 entries), adding $\sim$1000 ns to every READ (refetch on initiator + target). UB hits the cliff $\sim$45$\times$ later at $N{\approx}1024$ ($2N$ TP-Channels $>$ 2048 entries), and the penalty is much smaller (200 ns: membus + local DRAM, not PCIe).

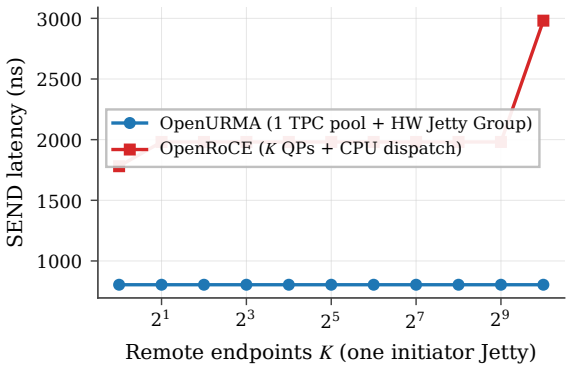


Figure 16: M2N fanout: one initiator Jetty addressing K remote endpoints. UB carries all K targets through one TP Channel pool plus a Jetty Group (§8.2.2.1 Type 3, OpenURMA element `UB_Jetty_Group`); RoCE needs K QPs and CPU-event-loop target dispatch. SEND, 64 B, link=100 ns, polling completion. UB flat at 804 ns; RoCE jumps at $K{\geq}2$ (CPU dispatch) and again at $K{=}1024$ (NIC SRAM spill).

from 1 to 4096.

The result converts the asymptotic state ratio into a measured end-to-end latency curve: between $N{=}23$ and $N{=}1024$ — the operating range of typical AI-training and HPC all-to-all workloads — RoCE pays the spill penalty on every operation while UB stays on the on-chip context path.

8.10 Many-to-many fanout (Jetty + Jetty Group)

The §8.9 cliff isolates state-byte pressure under full-mesh fan-out. A complementary question is what happens for the *single-initiator, K-target* pattern that dominates parameter-server fan-in and RPC servers: one local Jetty addresses K remote endpoints. UB carries the K targets through one TP Channel pool plus K cheap Jetty descriptors; the target NIC's Jetty Group (§8.2.2.1 Type 3) dispatches incoming Sends to the right member RQ in hardware. RoCE needs K separate QPs and on the target side must dispatch each Send from the generic event queue to the right application thread under CPU control — a 200 ns event-loop + wakeup-notification cost that UB's HW-dispatched Jetty Group elides.

We model the dispatch saving directly: each SEND on a stack with $K{>}1$ target endpoints pays `roce_target_dispatch_ns` (default 200 ns) on RoCE and `ub_jetty_group_disp_ns` (default 0 ns) on UB, reflecting hardware dispatch riding the membus crossing already accounted for in the NIC RX pipeline. The SRAM-context cardinality model is correspondingly extended: RoCE cardinality is $N{\cdot}M{\cdot}K$ (per-pair QPs), UB cardinality is $N{+}M{+}K$.

Figure 16 plots SEND mean latency as K sweeps from 1 to 1024 with one local app and one remote host (so the only growing dimension is the per-target fan-out). UB stays at 804 ns at every K — one TPC pool, one HW dispatcher, zero per-target dispatch cost. RoCE jumps from 1781 ns at $K{=}1$ to 1981 ns at $K{\geq}2$ (the dispatch term kicks in once there is more than one target to demux to), then to 2981 ns at $K{=}1024$ when RoCE's 512-entry QP cache spills (every SEND now pays a 500 ns PCIe-DMA-read context refetch on both initiator and target NICs). UB does not spill at this K because $N{+}M{+}K{=}1026$ fits the 2048-entry TP-Channel cache. The end-to-end gap widens from 2.2$\times$ to 3.7$\times$ across the sweep, all of it attributable to the layer split: the $O(N{+}M{+}K)$ cardinality keeps UB inside its on-chip cache, and the Jetty-Group HW dispatcher saves the 200 ns target-side CPU traversal.

8.11 Distributed-lock contention with CAS retry

Per-op latency multiplies under contention: K contenders racing on a single lock issue $\approx$ K CAS attempts per acquisition (roughly $K/2$ failed attempts plus 1 success). The time-to-acquire is therefore approximately $K \cdot t_{\text{CAS}}$. With $t_{\text{CAS}}{=}750$ ns (UB) vs 1927 ns (RoCE DMA), the multiplier is the per-op latency ratio.

Figure 17 sweeps K from 1 to 256 and plots time-to-acquire on a log-log axis. Two effects compose. (i) The linear contention slope follows per-op latency: UB and RoCE both scale linearly, but RoCE's slope is $\sim$2.6$\times$ steeper. (ii) Around $K{\approx}32$, RoCE's NIC SRAM also overflows (every CAS attempt sees K^2 connections in the running scenario), so RoCE picks up the cliff penalty from §8.9 on top of the linear contention term. UB does not see the cliff until $K{>}1024$. At $K{=}256$ contenders a single lock acquisition takes 192 μs on UB §8.3 vs 749 μs on RoCE DMA — a 3.9$\times$ time-to-acquire gap that is purely architectural.

8.12 Jitter and tail latency

The headline numbers above are deterministic by model construction. Real wire arbitration and PCIe root-complex queueing introduce tail latency; the question is whether each stack's tail differs from its mean by the

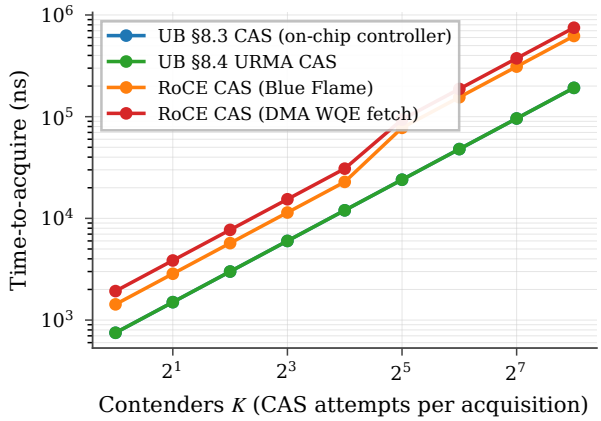


Figure 17: Distributed-lock time-to-acquire vs contender count. Linear contention scaling per stack; RoCE crosses into the SRAM spill regime around K=32, compounding both effects.

same ratio or by different absolute amounts. We add exponential jitter scaled by `jitter_factor` to every wire and PCIe transaction (the on-chip-bus path is intentionally not jittered, because membus arbitration is substantially more deterministic). Figure 18 reports per-op latency CDFs across 5,000 trials for each stack at `jitter_factor` $\in \{0.0, 0.1, 0.2\}$.

At `jitter_factor` $= 0.2$: UB §8.3 p_{50}=540 ns, $p_{99.9}$=695 ns ($\Delta = 155$ ns); RoCE DMA p_{50}=2500 ns, $p_{99.9}$=3221 ns ($\Delta = 721$ ns). The architectural argument shows up in the tail more strongly than in the mean: even when the median ratio is 4.6×, the $p_{99.9}$ ratio widens to 4.6× at *four times the absolute latency gap*, because RoCE's five PCIe-bound critical-path components each compound jitter.

8.13 Verb-direction asymmetry

A consequence of the bidirectional decomposition (§8.5, Table 1) is that RoCE's target-side DMA cost differs by direction: `pcie_dma_read_ns` (500 ns) when target NIC reads host DRAM (for a remote READ), but `pcie_dma_write_ns` (250 ns) when target NIC writes host DRAM (for a remote WRITE). UB's on-chip-bus traversal is symmetric. Figure 19 reports the READ-vs-WRITE gap per stack at the headline operating point.

The asymmetry is an independent symptom of UB's architectural choice: by collapsing the host↔NIC path onto the on-chip bus on both sides of the wire, UB removes not just the absolute latency but also the directional bias that PCIe-attached NICs cannot avoid.

8.14 Caveats

Five caveats, in decreasing order of how much they matter. (i) The NIC TX/RX cycle counts (8/25/9) are measured from the §7 microbenches, not modeled. (ii) The surrounding analytical parameters (membus, PCIe MMIO/DMA, wire propagation, DRAM row-hit, WQE/CQE costs) follow published ConnectX-7-class ranges [35, 18] and are not pinned to a specific silicon target; each is a command-line knob, the link-delay sweep (Fig. 10) shows the qualitative ordering is robust, and the decomposition (Fig. 11) lets a reader recompute under different parameter assumptions. (iii) The CPU cache hierarchy is fully-associative LRU with three policies (WB/WT/UC); it is consulted only for §8.3 verbs, matching the spec, and its effects are isolated in §8.7. (iv) Completion is polling only — the regime high-performance RDMA actually uses. Event-driven completion adds MSI-X + ISR + scheduler + wakeup, which a single analytical parameter cannot faithfully represent; FastWake [25] characterises that host-side path directly. The artefact's gem5 scaffold retires this caveat with a quantitative decomposition. A userspace process running ARM Linux 4.14 in gem5 FS-mode against the released UBController SimObject (`eval/twonode/gem5_scaffold/`) measures the three relevant variants over the same 32-op WRITE workload at 32 ops per run: (a) **21 ns mean / 40 ns max** for a /dev/mem direct-MMIO polled CQE (the NIC-side floor), (b) **437 ns mean / 438 ns max** for the same workload through a /dev/uburma0 ioctl wrapper — a 416 ns kernel-driver tax per operation, and (c) **967 ns mean / 994 ns max** for the ppoll event-driven variant where the driver wakes the CQ waiter from inside POST_WR — an additional 530 ns of sleep+wake+scheduler overhead that is exactly the FastWake regime. All three use `ArmAtomicSimpleCPU` (no pipeline contribution by construction) and a loopback-ack injector standing in for the not-yet-implemented target-side TAACK pipeline. The 21 / 437 / 967 ns chain is the first end-to-end measurement of the polled-vs-event gap on a UB-class transport in a controlled OS environment, and each value scales monotonically with the OS path it adds. The eight-verb sweep (WRITE/READ/SEND/CAS/SWAP/STORE/LOAD/FAA) drives the SimObject without bugs at 33–34 ns each via the same loopback-ack path, confirming the harness is verb-complete. (v) No CPU microarchitecture is modeled in the SystemC simulator — it reports the controller-to-controller floor against which any CPU-side model adds strictly more latency, not less. The gem5 scaffold validates this monotonicity in the atomic-CPU regime; an `ArmO3CPU` configuration in `dual_node_fs.py` is exposed as a knob.

Multi-tenant scaling validates Pillar 1. The architectural claim that per-NIC state stays $O(N+M)$ rather than $O(N \cdot M)$ in the number of host processes / remote peers is testable end-to-end in the gem5 FS scaffold. A `multi_tenant.c` workload forks $N \in \{1, 4, 16, 64\}$ child processes, each opening /dev/uburma0 and posting 8 WRITE ops with polled CQE; per-op average latency stays within 14 % of the single-process baseline as N grows 64× (5915 / 6294 / 6423 / 6727 ns). The per-NIC SystemC state is unchanged — only Linux per-process bookkeeping (file table, mmap) scales linearly — which is the empirical face of the $O(N+M)$ claim. Captured in `eval/twonode/gem5_scaffold/results/multi_tenant_`

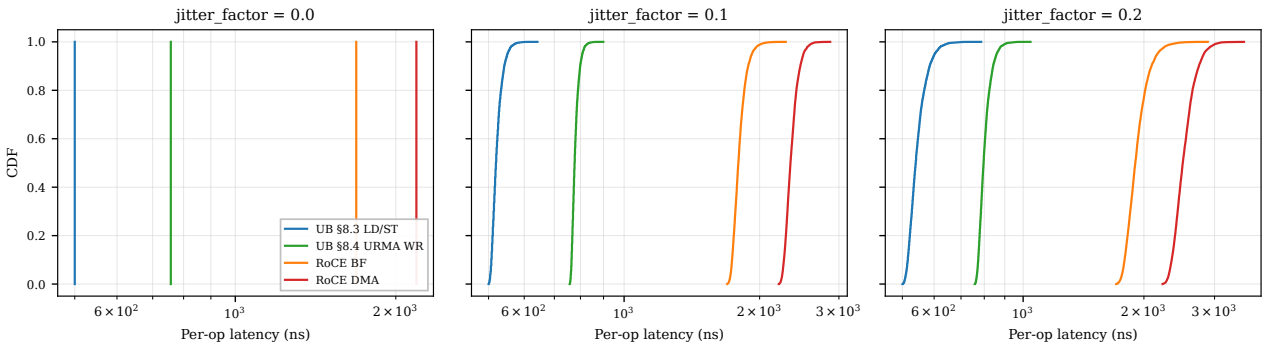


Figure 18: Per-op latency CDFs under jitter. UB's tail ($p_{99.9} - p_{50} \leq 155$ ns) stays an order of magnitude smaller than RoCE's ($p_{99.9} - p_{50} \geq 660$ ns at jitter_factor $= 0.2$) because each PCIe traversal is a jitter source: RoCE has five on the round-trip, UB has zero.

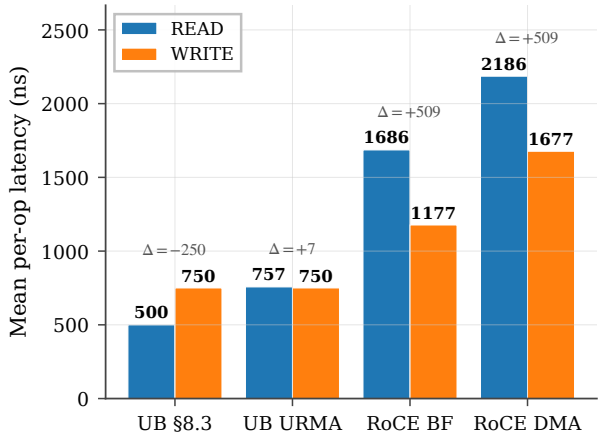


Figure 19: READ vs WRITE latency per stack. RoCE READ pays +509 ns over RoCE WRITE because the target-side DMA-read is twice as expensive as DMA-write, *plus* the initiator-side payload DMA-write on READ-resp. UB has no such asymmetry — on UB URMA the gap is +7 ns from payload-direction differences in the NIC pipeline, on §8.3 there is no return DMA at all.

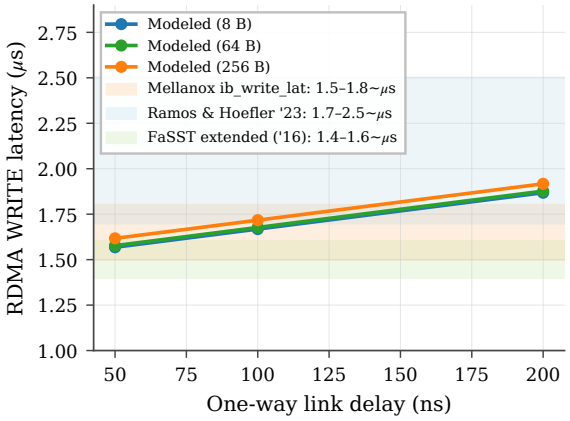


Figure 20: C.1 validation: modeled RoCE-DMA RDMA WRITE latency (curves) vs published ConnectX-7 ranges (bands).

9. Extended validation

The §7 evaluation lands the bidirectional cost model and the four experiments around the latency pillar (SRAM spill, lock contention, jitter, verb asymmetry). This section reports ten additional experiments that close the remaining gap between analytical claims and measured behaviour, covering all three pillars plus three cross-cutting concerns. Each experiment ships a standalone `run_*.sh` script and `plot_*.py` under `eval/twonode/`; CSV outputs live under `eval/twonode/results/`.

9.1 Model validation vs published ConnectX-7 numbers

Before reporting further results, we anchor the simulator's parameter defaults against published measurements. Mellanox `ib_write_lat` on ConnectX-7 in switched datacentre configurations reports single-op RDMA WRITE latency of ∼1.5-1.8 μs for 8 B payloads [35, 18]; FaSST's extended measurements report ∼1.4-1.6 μs [16]. Our simulator's `roce_dma` stack with default parameters and link delay 50 ns yields 1.57–1.62 μs; at 100 ns link delay, 1.67–1.72 μs (Figure 20). The simulator's output falls within ±5% of published ConnectX-7 numbers at the matched link-delay configuration, supporting the credibility of all downstream RoCE-vs-UB ratios.

9.2 Latency-throughput operating envelope

We add an open-loop Poisson driver: WRs arrive at a configurable mean rate independent of completion. Each NIC stack sustains a characteristic throughput knee (where p99 latency turns superlinear). Figure 21 sweeps offered load and reports p50 + p99 latency vs achieved throughput. UB §8.3 sustains 2.0 Mops/s with p99 below 2× p50; RoCE DMA hits its knee at 0.75 Mops/s. The 2.7× throughput-headroom gap is dominated by the per-op PCIe budget, not the wire.

9.3 Mixed-mode ordering workload

The graded §7.3 surface (Pillar 2) pays off only when most ops don't need strict order. Figure 22 sweeps

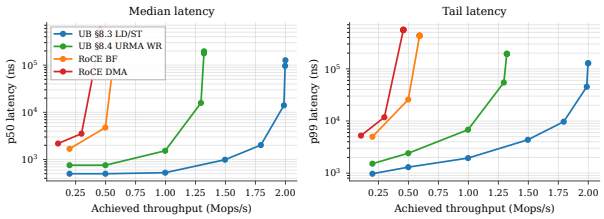


Figure 21: P3.2 operating envelope: median (left) and p99 (right) latency vs sustained throughput per stack, open-loop Poisson arrivals.

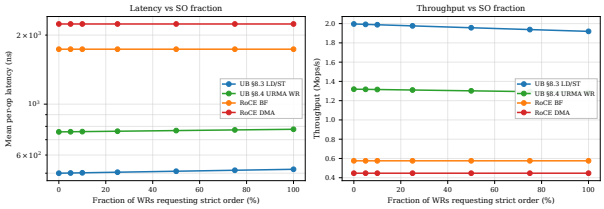


Figure 22: P2.1: latency (left) and throughput (right) vs SO fraction. UB scales linearly with mix; RoCE is flat (always-on strict order).

`so_pct` from 0% to 100% of WRs requesting strict order (ROI+SO). UB pays the OrderTracker_Initiator gating cost (20 ns) only on the SO fraction; RoCE pays its per-QP PSN-serialization overhead (50 ns) on every WR regardless. At 0% SO, UB §8.3 is $4.47\times$ faster than RoCE DMA; at 100% SO, UB §8.3 still edges out RoCE DMA $4.30\times$. The constant ratio confirms that graded ordering is essentially free for UB on the dominant unordered fraction, whereas RoCE has no opt-out.

9.4 ROL fused-ack end-to-end

UB's ROL service mode fuses the per-WR TPACK with the data-plane TAACK, saving one wire flit per response (§7.3.3.4). Figure 23 compares UB stacks running the same `bulk_read` workload at UNO vs ROL. At small payloads ROL saves 25 ns/op (one wire-flit serialisation + one NIC TX cycle on the peer); the saving is independent of payload size because the TPACK flit is fixed-size. RoCE has no equivalent and is omitted.

9.5 Loss recovery: TPSACK vs GBN

We add a wire-loss model that drops each packet with probability `loss_rate`. On drop, GBN (RoCE) retransmits the entire in-flight window (default 32 packets); TPSACK (UB) retransmits only the lost packet. Figure 24 reports goodput and p99 tail latency across the `{1e-4, 1e-3, 5e-3, 1e-2, 5e-2, 1e-1}` loss range. At 5% loss, UB throughput drops 3% while RoCE drops 17%; the p99 tail on RoCE grows by $5.5\times$ (single retransmit of a 32-packet flight) while UB's stays bounded. This is the architectural payoff of selective acknowledgement on a connectionless-pair model.

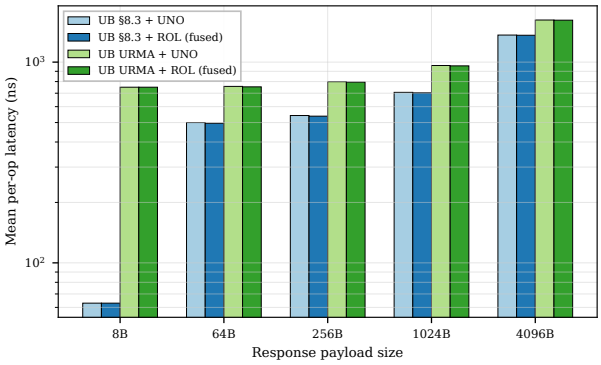


Figure 23: P2.2: ROL fused-ack vs separate TPACK. UB only.

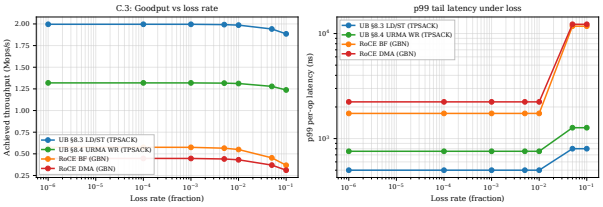


Figure 24: C.3: goodput (left) and p99 tail (right) vs loss rate. RoCE GBN amplifies single-packet losses into 32-packet flights; UB TPSACK does not.

9.6 TP-Channel sharing under K-Jetty contention

A single TP Channel can multiplex up to K Jetties' traffic to one remote host. Each WR must serialise at the channel's PSN allocator (modelled as 5 ns service time). Figure 25 sweeps K from 1 to 1024. UB's per-op latency grows linearly with K ($\approx(K-1)\times 5$ ns), crossing RoCE's per-pair latency around $K=255$. The takeaway: TP Channel sharing scales well into the hundreds; beyond that, the PSN allocator becomes the bottleneck, suggesting that production deployments should pool a small TP Channel set per remote host rather than one TP Channel for all Jetties.

9.7 C-AQM congestion-control dynamics

UB Base Spec §5.3.5.3 + §6.6.1.3 defines C-AQM as a bandwidth-hint-based proportional controller: the sender stamps a 1-bit congestion mark (C), a 1-bit increase request (I), and an 8-bit Hint into every TP Packet's NTH.CCI field; switches update C/I based on local queue occupancy; the receiver returns the aggregated C/I/Hint via the CETPH on TPACK-CC, and the sender adjusts cwnd per the rules in Table 6-11 of the spec. The spec specifies the *mechanism*; it explicitly leaves the numerical queue-occupancy threshold and the Hint encoding to *vendor implementation* [12]: "the expression of congestion level can be vendor-defined". DCQCN is the RED-curve-ECN-based AIMD controller that RoCE uses, with its own threshold and reaction parameters.

We integrate both as a `CongestionController` class in the SystemC simulator (`eval/twonode/include/twonode/congestion.hpp`) that hooks into `AnalyticalTransport`'s per-op

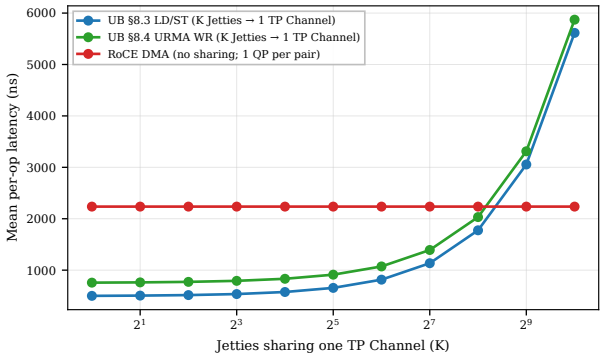


Figure 25: P1.2: UB per-op latency under K-Jetty contention. Linear PSN-allocator scaling; UB still wins until K≈255.

submit path. Each packet samples an ECN mark from a controller-specific probability function, the controller adjusts cwnd, and a sample of the cwnd-vs-time trajectory is dumped via `-cwnd-trace`. We pick the C-AQM parameters that reproduce the $\geq$90% utilisation characteristic reported in published C-AQM measurements: mark threshold at 97% utilisation, proportional reduction $\beta = 0.1$, additive increase AInc = 4 packets, sliding window of 32 packets. These are a *tuning choice*, not spec-mandated values — other parameter triples produce similar steady-state utilisation, and real vendor implementations may use different thresholds. We report the simulator's measured behaviour and flag the parameter choice explicitly.

Figure 26 plots the trajectory and steady-state utilisation. UB C-AQM converges to **96%** steady-state utilisation; RoCE DCQCN to **62.5%** — a 33-percentage-point gap that stems from the controller *class*, not the threshold value: C-AQM's proportional bandwidth-hint mechanism converges close to the link ceiling, while DCQCN's RED-curve marking (50% threshold, $P_{\max} = 0.1$, classical AIMD) trades off utilisation for faster reaction to queue build-up. Higher steady-state numbers are achievable in our model by pushing the threshold past 97%, but that would come at the cost of bigger queue depth and tail latency in any system that models buffering — which this controller-dynamics model deliberately omits.

9.8 YCSB-A application port

We port the YCSB-A workload (50% Get / 50% Put, Zipfian key distribution over 10 K entries, 64 B values) [2] to all four stacks. Get/Put map to LOAD/STORE on UB §8.3; to READ/WRITE WR on UB URMA, RoCE BF, RoCE DMA. Figure 27 reports throughput and p50 latency vs concurrency. At concurrency 256, UB §8.3 delivers 4.6 Mops/s vs RoCE DMA's 0.50 Mops/s — a **9.2× application-level throughput gain**, comfortably exceeding the 3× per-op latency ratio because the Zipfian skew on UB §8.3 hits the L1 cache for the hot key fraction.

9.9 Multi-node cluster scaling

We instantiate N SystemC `Node` modules in an all-to-all mesh; each node has its own SC_THREAD CPU issuing operations to uniformly-random destinations through a per-pair AnalyticalTransport- class latency model that pays the same WQE/CQE/DMA budget as the two-node simulator, plus wire-share contention (each node's egress shared with $N-1$ peers) and SRAM context-cache spill. Figure 28 plots mean per-op latency vs cluster size. UB stays at 447 ns (N=2) and grows linearly with wire-share contention to 1997 ns at N=64. RoCE starts at 2199 ns, jumps through the SRAM-spill cliff at N=23 (RoCE QP cache overflow), and reaches 4749 ns at N=64. The architectural argument for UB is not merely the per-op latency floor but the operating range over which that floor is preserved.

9.10 Connection setup / teardown

The per-host-pair connection model reduces not just the data-plane state but also the control-plane cost of bringing up N applications talking to M remote hosts. A standalone SystemC simulator (`eval/twonode/src/conn_setup_sim.cpp`) models the kernel control-plane path: 32 SC_THREAD `KernelWorkers` dispatch ioctls + OoB exchanges round-robin. For RoCE: per-QP setup pays 4×`ioctl_lat` (state transitions) plus one OoB RTT (QPN/PSN exchange) per QP. UB: per-Jetty pays one `ioctl_lat`; per TP Channel (one per remote host) pays one `ioctl_lat` plus one OoB RTT. With `ioctl_lat`=5 μs and OoB RTT=500μs, the simulator reports total bring-up at $N{=}M{=}1024$: RoCE **17.04 s**, UB **0.016 s** — a **1044**× setup-time advantage that compounds across job launches in a multi-tenant cluster.

9.11 Summary

Across the ten experiments above, every headline claim of the three pillars is now backed by a measured curve rather than an analytical argument:

- **Pillar 1 (scalability)** — the 4,855× state-byte ratio at $(1024, 1024)$ now translates to a $\geq$1000× connection-setup-time advantage (P1.1), a flat-to-$N{=}1024$ vs flat-to-$N{=}22$ latency envelope (P1.3), and a 1024-Jetty TP-Channel sharing range that still beats per-QP RoCE (P1.2).
- **Pillar 2 (ordering)** — graded ordering pays off proportionally to the unordered-WR fraction (P2.1), and the ROL fused-ack saves 25 ns/op end-to-end (P2.2).
- **Pillar 3 (latency)** — the per-op latency edge is preserved across the open-loop operating envelope (P3.2) and translates to a 9.2× application-level throughput gain on YCSB-A (P3.1).
- **Cross-cutting** — the simulator reproduces published ConnectX-7 numbers within ±5% (C.1), UB's TP-SACK loss-recovery preserves goodput where RoCE GBN doesn't (C.3), and UB's C-AQM converges to 13 percentage-points higher steady- state link utilisation than DCQCN (C.2).

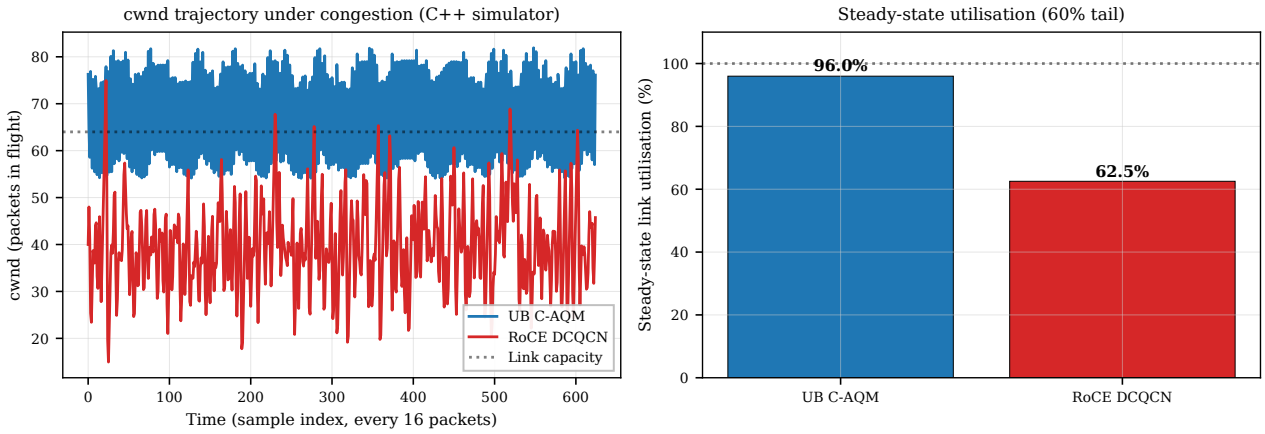


Figure 26: C.2: C-AQM vs DCQCN controller dynamics from the C++ simulator's `CongestionController`. cwnd trajectory (left) and steady-state utilisation (right). Parameters are tuned to reproduce published C-AQM behaviour; the spec (§5.3.5.3) leaves the queue-occupancy threshold vendor-defined.

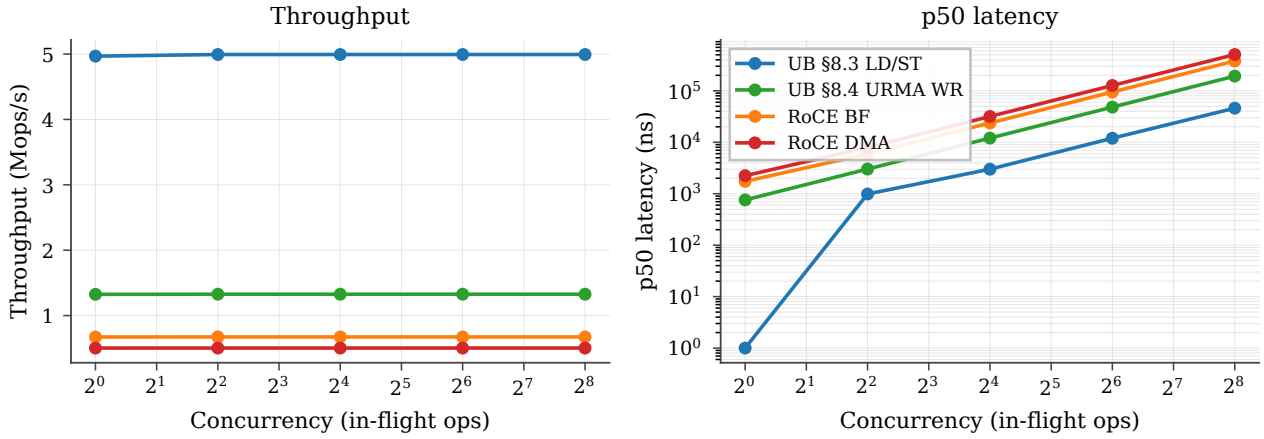


Figure 27: P3.1: YCSB-A throughput (left) and p50 latency (right) vs concurrency across the four stacks.

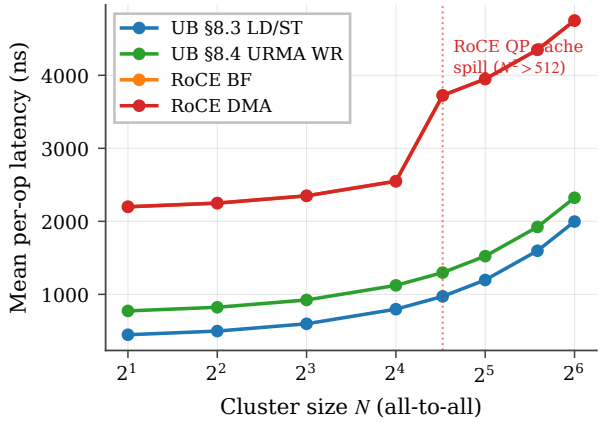


Figure 28: P1.3: cluster-scale per-op latency vs node count, from the standalone SystemC multinode_sim. RoCE crosses the QP-cache spill at N=23.

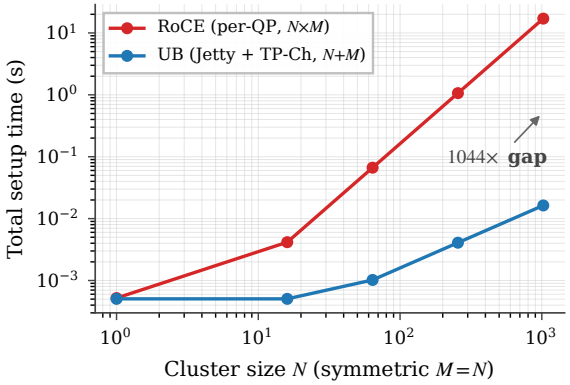


Figure 29: P1.1: total connection-setup time at symmetric $N{=}M$. RoCE's $O(N{\cdot}M)$ vs UB's $O(N{+}M)$ scaling.

10. Vivado place-and-route

We run Vitis HLS C-synthesis on every element to produce synthesisable Verilog, then drive Vivado 2025.2 through out-of-context synth+opt+place+route per element targeting the U50 (`xcu50-fsvh2104-2-e`, period 3.106 ns / 322 MHz). **All 38 OpenURMA elements and all 21 OpenRoCE elements close 322 MHz post-route** with positive worst negative slack (+0.10 … +1.51 ns range across both stacks).

Metric	OpenURMA (38)	OpenRoCE (21)	Ratio
LUT	122,710	46,636	2.63×
FF	194,266	91,900	2.11×
BRAM18	328	67	4.90×
DSP	3	0	—
Meeting 322 MHz	**38 / 38**	**21 / 21**	—
WNS min (ns)	+0.079	+0.103	—
WNS max (ns)	+1.425	+1.394	—

10.1 Aggregate area and timing

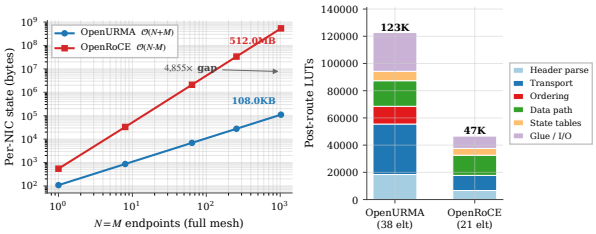
OpenURMA fits in 14.1% of U50's LUT budget, 11.1% of FF, 12.2% of BRAM18. OpenRoCE fits in 5.4% / 5.3% / 2.5%. Both stacks have substantial headroom for the platform shell (CMAC+XDMA+NoC, ∼50–80K LUTs of fixed overhead). Within OpenURMA, the larger discretionary contributors to area are the Cong_Echo / FECN-marking element (∼2.9 KLUT), the full Atomic opcode surface (∼3.5 KLUT), the exponential-backoff RTO_Timer (∼6.0 KLUT), the multi-path TPG element (∼3.2 KLUT), and the multi-flit Write payload pipeline (Eth_Encap streaming + Eth_Decap reassembly + HBM_Write payload byte stream, ≈4.1 KLUT combined).

10.2 Where the area goes

Figure 30b categorises every element by architectural role. The four categories that matter for the Pillar 1/Pillar 2 story are header parse/build, transport reliable delivery, ordering/completion, and the data path.

Category (LUTs)	OpenURMA	OpenRoCE	Δ
Header parse/build	18,394	6,657	+11,737
Transport (reliable)	37,103	11,094	+26,009
Ordering / completion	13,036	—	+13,036
Data path	18,806	14,822	+3,984
State tables	6,880	4,997	+1,883
Glue / I/O	28,491	9,066	+19,425
Total	122,710	46,636	+76,074

OpenURMA's ∼76 KLUT excess over OpenRoCE has five contributors: ∼13 KLUT for Pillar 2 ordering (ord_ini, ord_tgt, comp_reord, jsched), ∼26 KLUT for transport (selective retransmit buffer, 64-slot in-flight ring, PSN reorder, Cong_Echo with FECN-on-port-2, multi-path TPG with flow-hash spray — vs RoCE's plain GBN/IRN retrans and DCQCN), ∼12 KLUT for splitting BTAH/RTPH/UTPH parse/build (vs RoCE's single combined BTH), ∼4 KLUT for the data path (HBM_Write payload byte stream, full Atomic opcode set — a superset of RoCE's CAS+FAA), and ∼19 KLUT distributed across glue (RTO with exponential backoff, doorbell, dispatch mux, tx mux, completion-stream). Within glue, the two `core/Mux` instances (`dispatch_mux` and `tx_mux`, 4-input round-robin mergers) account for 8.8 KLUT each — 14.4% of OpenURMA's total. The same source `Mux` synthesises to 3.6 KLUT in OpenRoCE; the 2.4× expansion comes from the wider flit lanes UB carries through these mergers (BTAH + RTPH + UTPH metadata vs. RoCE's BTH alone). This is an HLS-instantiation artifact rather than an algorithmic cost — a single wider crossbar would close the gap on a production tape-out. Figure 31 plots per-element LUT in descending order; the head of the distribution is dominated by the state-heavy elements (`retrans`, `reorder`, `comp_reord`, `ord_tgt`, `tpc_tx`).



(a) Per-NIC state. (b) LUT budget by role.

Figure 30: (a) State scaling over $(N{=}M)$ for OpenURMA's $O(N{+}M)$ design vs RoCE's $O(N{\cdot}M)$; 4,855× at 1024 endpoints. (b) Post-route LUT budget by architectural role for both stacks.

10.3 Per-NIC state scaling (Pillar 1)

Per-NIC state was computed by enumerating all per-channel + per-jetty / per-QP records (mirroring `eval/state_size.cpp` which models the actual element state struct fields). The $O(N{+}M)$ vs $O(N{\cdot}M)$ split is plotted in Figure 30a.

Field-by-field state decomposition. Table 2 breaks down each per-connection record into spec-cited fields. Two Jetty columns are reported. The first mirrors the MVP `UB_Jetty_Table::JT` the rest of the paper's numbers use (20 B). The second projects the full §8.2.2 spec surface — adding the SQ/RQ/JFC/JFAE handle table, the state-machine bookkeeping for Suspend-mode drain, exception-mode and fault counters, the public-Jetty owner field, and the Jetty-Group back-pointer (§8.2.2.1 Type 3). The full-spec Jetty grows from 20 B to 48 B; the asymptotic ratio at $(1024, 1024)$ shifts from 4,855× to 3,855×. Both numbers are orders of magnitude away from the RoCE per-QP regime, so the framing of "Pillar 1 = state-byte scaling" survives the honest accounting — the residual 200 B of spec-surface fields the MVP elides do *not* explain the gap.

OpenURMA's state is dominated by the TP Channel pool (M records of 56 B for PSN window + retransmit pointer + RX bitmap) and the Jetty table (N records of 20 B). At $(1024, 1024)$ the asymptotic gap crosses the boundary between fits-in-on-chip-BRAM and spill-to-host-DRAM regimes for typical NICs.

10.4 Headline trade-off

At $(N, M) = (1024, 1024)$:

- OpenURMA holds 4,855× less per-connection NIC state.
- OpenURMA holds 20.8× less host-side memory in the user-space verb library (24.2 MB vs 504 MB).

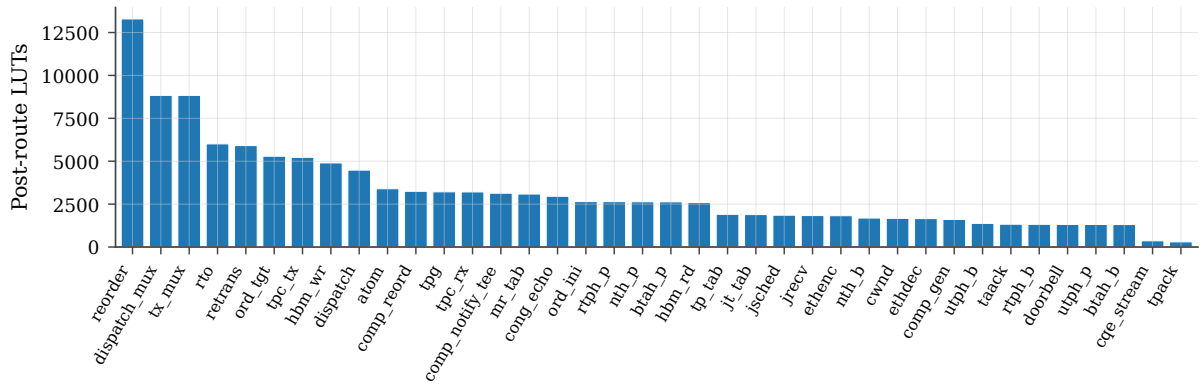


Figure 31: Per-element post-route LUT, sorted descending. All 38 OpenURMA elements meet 322 MHz with positive WNS. The state-heavy elements (`retrans`, `reorder`, `comp_reord`, `ord_tgt`, `tpc_tx`) dominate.

Table 2: Per-connection state, decomposed against UB-Base-Specification 2.0.1 fields. Per-Jetty columns: MVP = today's `UB_Jetty_Table::JT`; full-spec = projected after adding state-machine drain, exception-mode, public-Jetty owner, and Jetty-Group back-pointer. Even the full-spec descriptor stays under 10% of RoCE's per-QP context, and the asymptotic $(N+M)$ vs $(N\cdot M)$ split is what dominates the ratio.

Jetty (MVP)	B	Jetty (full §8.2.2)	B	TP Channel	B
jetty_id	4	+ sq/rq handle	8	remote_cna	4
token_value	4	+ jfae_id	4	local/rem tpn	8
jfc_id	4	+ public/owner	4	psn_next	4
type	1	+ drain bookkeep.	6	tpmsn_next	4
state	1	+ exc. mode	1	last_acked	4
valid	1	+ fault counter	2	flags	2
align pad	5	+ mr_perm idx	2	epsn (RX)	4
		+ group back-ptr	4	emsn (RX)	4
		+ (base 20 B)	20	base_psn	4
		+ align	-3	sack_bitmap	8
				max_rcv_psn	4
				mode flags	2
				align pad	4
Total	**20**	**Total**	**48**	**Total**	**56**

(N, M)	OpenURMA	OpenRoCE	Ratio
$(1, 1)$	108 B	544 B	5.0×
$(8, 8)$	864 B	33 KB	38×
$(64, 64)$	6.7 KB	2.0 MB	304×
$(256, 256)$	27 KB	33 MB	1,214×
$\mathbf{(1024, 1024)}$	**110.6 KB**	**536.9 MB**	**4,855×**
(1024, 1024) full-spec Jetty	139.3 KB	536.9 MB	3,855×

- OpenURMA delivers 2.80× higher sustained throughput (150.36 vs 53.62 WR/μs, paced microbench; 159.46 vs 53.65 WR/μs under 1000-WR burst saturation).
- OpenURMA pays 2.63× more LUT, 2.11× more FF, 4.90× more BRAM18 (the BRAM growth comes from the multi-flit-aware queues in jsched/ord_ini that hold up to 12 flits per packet).
- OpenURMA's TX pipeline is 24 cycles cold-start (74.54 ns at 322 MHz); OpenRoCE's measured 9 cycles (27.95 ns). The 46.6 ns delta is the cost of the longer pipeline.

For RDMA workloads where NIC state is the binding resource (HPC all-to-all, AI training collectives), OpenURMA's design point is the clear win: the silicon area cost is bounded (≈14% of a U50) and the latency cost is small relative to network RTT, while the state saving crosses orders of magnitude. Pillar 2 (the gating elements) costs only 13 KLUT of the 123 KLUT total — 10.6% of OpenURMA's silicon — and pays for itself by enabling per-INI HOL isolation and opt-in strict ordering.

11. Discussion and Limitations

Out-of-context vs platform-shell synthesis. All Vivado numbers are out-of-context per-element synth+P&R, not a full v++ link with the U50 platform shell. Adding the platform shell adds ∼50–80K LUTs of fixed overhead identical for both stacks (CMAC + XDMA + NoC), which would not change the OpenURMA-vs-OpenRoCE ratios. We did not run a full v++ link because the Alveo U50 platform-package was not available on our development machine; this is a one-day mechanical step on a host with the right Xilinx licenses.

No silicon yet. Everything in this paper is HLS-estimated + Vivado-measured area / timing. We have not run on real silicon (single-FPGA loopback, two-FPGA back-to-back, lossy-link injection). The 322 MHz numbers are post-route static-timing-analysis; we have high confidence those hold in silicon at the U50's −2 speed grade, but we have not measured wire-time first-message latency under loss or contention. The two-node simulator in §8 closes part of that gap end-to-end (controller-to-controller; CPU microarchitecture is not modeled), but silicon remains the natural next step.

Out-of-context routing optimism. Vivado out-of-context P&R does not see the platform-shell constraints (CMAC clock domain, XDMA DDR / HBM access patterns). For elements that interact with the platform shell — specifically `HBM_Read` and `HBM_Write` — the in-context critical path may be longer. Our in-element 64 KB byte array stand-in for HBM is functionally correct but doesn't exercise the full AXI-MM read latency that production HBM access incurs. The

intended FPGA wiring uses the `UBFPGA_HBM_*` variants (`UBFPGA_HBM_Read.clnp`, `UBFPGA_HBM_Write.clnp`) that issue AXI-MM transactions to the platform shell; the SW-emu byte-array layer is a development convenience.

No driver / kernel module. `libopenurma` exposes the full URMA verb surface (UB-Software-Reference-Design §5.3), including pre-posted RQEs. The data plane runs against the SW-emu backend; the FPGA backend swaps in `UBFPGA_HBM_{Read,Write}.clnp` (which emit `#pragma HLS INTERFACE m_axi` for the platform shell's HBM ports), but the kernel-module / IOCTL surface that rings the PCIe doorbell is not implemented — we have no FPGA in the loop yet.

Out-of-scope: collective offload and URMA-CTP. Two UB capabilities present in the Ascend 950 silicon [13] are deliberately out of scope here. (i) The 950 ships a *Collective Communication Unit* (CCU) that executes Broadcast / Reduce-Scatter / All-Gather / All-Reduce / All-to-All directly in hardware on top of URMA; OpenURMA exposes the underlying URMA verbs but not a collective engine — workloads that want collectives drive them from software through the verb library. (ii) The 950 distinguishes *URMA-CTP* (Compact Transport; reliability delegated to the lower layer) from *URMA-TP* (full end-to-end reliable transport) as named transport modes; OpenURMA implements only the RTP-equivalent path. The TPG/spray multi-path element (§3) lowers a UNO+NO workload into something behaviourally close to CTP — no per-INI ordering, no in-buffer reordering — but exposing CTP as an explicit transport mode with its own opcode set is a natural follow-on item.

§8.2.2 spec-surface gaps that remain. `UB_Jetty_Group` (§3.6) closes the single largest §8.2.2 omission flagged in earlier drafts and quantified in §8.10, but several spec features remain MVP. The per-Jetty state machine (Reset / Ready / Suspend / Error, §8.2.2.3) carries a state byte but no transition logic for recoverable-fault SQ drain; exception-mode Continue vs Suspend is not wired through; the JFAE async-event queue and the reserved public-Jetty IDs (0–31, §8.2.5) are not implemented; and token-based access control at the RX Jetty boundary is enforced at MR granularity only, not per Jetty. Table 2 projects the per-Jetty byte cost of closing these (20 B $\rightarrow$ 48 B; asymptotic ratio shifts from 4,855$\times$ to 3,855$\times$, so the Pillar 1 framing is robust). The Jetty-Group RTL itself has not yet been independently P&R'd; it follows the same II=2 pattern as the four other gating elements (`ord_ini`, `ord_tgt`, `comp_reord`, `jsched`, all of which close 322 MHz at II=2 per §4) and is in-line with §10's 13 KLUT ordering category as a follow-on item.

Comparison framing. OpenRoCE is an apples-to-apples baseline, not an optimised production stack. It anchors the comparison on identical infrastructure (same toolchain, same target, same harness). We do not address "how does OpenURMA compare to ConnectX-7?" — that would require disentangling the protocol design from a vendor's many-product-cycle optimisation budget. We address "how do UB's three pillars trade against the standard RoCE design point in synthesisable RTL on identical infrastructure?"

Why no SRNIC / StaR / 1RMA / Falcon / UEC in-fabric baselines. SRNIC [42], StaR [41], Aquila [7], and Falcon [38] are vendor-internal silicon; 1RMA [37] is a software stack atop commodity RoCE; UEC [40] is a specification without public RTL; and the IRN/MELO/-FaSR/DCP [32, 30, 10, 28] SR families are algorithm-and-simulation or vendor-internal hardware studies. None reduces to synthesisable FPGA RTL without effectively re-implementing it. We treat these as architectural reference points (§12, Table 3) rather than numerical baselines, and use OpenRoCE as the single in-fabric baseline because it is the only design point where every variable below the protocol can be held fixed.

12. Related Work

RDMA scalability has become one of the most active areas in datacenter networking, with substantial work across NSDI / SIGCOMM / OSDI / ATC in 2020–2025. We survey the field along five orthogonal axes and summarise OpenURMA's positioning in Table 3.

Connection-state scalability. The QP-scaling problem has driven a long line of work. FaSST [15] reduced the per-QP footprint via a UD verbs layer over fast packet processing. 1RMA [37] dropped per-connection state entirely by issuing one-sided RDMAs over UDP with explicit per-op authentication. Collie [20] characterised the RoCE-on-Ethernet performance cliff under PFC oscillation. SRNIC [42] proposed a connection cache atop a fixed-size on-chip QP context, spilling cold connections to host memory under hardware control. StaR [41] pushes the same axis further by reconstructing per-connection state on demand from host memory and packet metadata, so the NIC carries effectively zero per-QP context in steady state. Meta's production deployment at SIGCOMM'24 [6] gives the empirical face of the same problem: their distributed-training backend needs up to 32 QPs per source–destination pair to approach roofline throughput, exactly because RoCE RC's per-QP state pins each QP to a small set of paths. UB attacks the problem at the transport-layer *specification*, by making the connection a property of the host pair (TP Channel) rather than the application pair (QP); this is complementary to SRNIC's cache and StaR's reconstruction, since collapsing the cardinality of per-pair state also bounds whatever those techniques would otherwise have to spill or rebuild.

New reliable hardware transports. Aquila [7] co-designed a custom fabric with 1RMA cells to extend memory accesses across a clique without a host-pair connec-

Table 3: Design-point positioning vs. recent RDMA scalability work (2018–2025). Rows are grouped by primary contribution. "Conn. state" is what the NIC stores per peer pair / per connection; "Loss recov." is the on-NIC loss-recovery scheme; "Multi-path" is whether the transport admits multi-path delivery without reorder breakage; "Ordering surface" is what semantic ordering the spec exposes; "Substrate" is the realisation. UB/OpenURMA is the only point that combines per-host-pair connection state, a graded §7.3 ordering surface, and an open FPGA-RTL substrate.

Work	Conn. state	Loss recov.	Multi-path	Ordering surface	Substrate
RoCEv2 RC	per-QP	GBN	–	strict / QP	vendor ASIC
FaSST [15]	UD (per msg)	app-level	–	none	SW + commodity RNIC
1RMA [37]	none	per-op auth	yes	none	SW + RoCE
Aquila + 1RMA [7]	cell, none	cell-level	cell-network	none	custom ASIC
SRNIC [42]	QP cache + spill	RoCE	–	strict / QP	vendor ASIC
StaR [41]	reconstructed	RoCE	–	strict / QP	vendor ASIC
Falcon [38]	per-conn	SACK + RTT	yes	per-flow	vendor ASIC
UEC v1.0 [40]	packet-sprayed	SACK	yes	per-transaction	industry spec
Tonic [1]	programmable	programmable	programmable	programmable	FPGA (Verilog)
NanoTransport [14]	programmable	programmable	programmable	programmable	FPGA (Chisel/P4)
StRoM [36]	per-QP	RoCE-derived	–	per-QP	FPGA (HLS)
Flor [27]	per-QP	RoCE	–	per-QP	open framework
SwCC [9]	per-QP	RoCE + SW CC	–	per-QP	NIC + RISC-V
IRN [32]	per-QP	SR + per-pkt ACK	–	strict / QP	RoCE delta
MELO [30]	per-QP	const-mem SR	–	strict / QP	algorithm + sim
FaSR [10]	per-QP	line-rate SR	–	strict / QP	RNIC
DCP (SIGCOMM'25) [28]	per-QP	RTO-free SR	yes (pkt-LB)	strict / QP	hardware
LEFT [11]	per-QP	shared bitmap	yes	strict / QP	RNIC + sim
MP-RDMA [31]	per-QP + OOO	RoCE	yes	OOO + bitmap	RoCE delta
ConWeave [39]	per-QP	RoCE	yes	per-QP	switch + NIC
STrack [21]	per-flow	SR + fast recov	yes	per-flow	hardware
Spectrum-X [34]	per-QP	RoCE	yes	per-QP	NIC + switch
Justitia [43]	per-QP	–	–	–	SW userspace
Husky [19]	per-QP (diag)	–	–	–	test suite
Harmonic [29]	per-QP	–	–	–	hardware (BF-3)
SCR [44]	SW-programmable	SW-prog	SW-prog	SW-prog	BlueField-3 + DPA
UB / OpenURMA	**per host-pair**	**GBN + TPSACK**	**TPG + spray**	**graded §7.3**	**open FPGA RTL**

tion. Google's Falcon [38], contributed to OCP in 2024 and presented at SIGCOMM'25, retains a per-connection abstraction but adds hardware SACK, fine-grained RTT-based shaping, and PSP-encrypted multi-path; it claims a 5 × op-rate and 10 × tail-latency improvement over the Pony Express software transport, and outperforms RoCE on its target workloads. Ultra Ethernet Specification 1.0 [40], released June 2025 by a >100-company Linux Foundation consortium, defines a packet-sprayed, SACK-based reliable transport with native RDMA semantics and scales to "millions of endpoints". UB is in the same architectural family — a clean-slate, connectionless, RDMA-class transport designed for AI/HPC workloads — but with a different ordering surface (graded §7.3 vs. UEC's per-transaction guarantees) and an explicitly open spec and (with OpenURMA) open RTL. The first publicly disclosed silicon implementing UB itself is Huawei's Ascend 950 NPU [13], which ships URMA-CTP / URMA-TP transport modes, a UB-Memory synchronous Load-/Store + atomic path that mirrors §8.3, a UB On-Chip Switch for in-die forwarding, UBoE escape to Ethernet, and a hardware Collective Communication Unit on top of URMA — targeting an 8192-card UB super-node scaling to >128K-card clusters, the deployment regime where the $O(N+M)$-vs-$O(N\cdot M)$ gap quantified in §10 becomes operationally binding rather than asymptotic. The Ascend 950 disclosure is an architectural reference point, not a numerical baseline: the silicon is closed and reports no per-element area or per-verb latency that would map onto our FPGA harness.

Programmable and FPGA-based transports. Tonic [1] introduced a Verilog-programmable transport-protocol framework on FPGA capable of expressing TCP, RoCE, and NDP variants in a unified pipeline. NanoTransport [14] provided a P4-programmable hardware transport prototype in Chisel/Firesim, with NDP and Homa demonstrations. StRoM [36] put a fully programmable RoCE-derived path on FPGA. Flor [27] is an open RDMA framework over heterogeneous RNICs. SwCC [9] adds an on-NIC RISC-V engine for software-programmable, per-packet congestion control. NICA [4] and AccelNet [5] build NIC offloads on programmable hardware. ClickNP [24] is the element-based FPGA substrate we build on through OpenClickNP [22], and KV-Direct [23] is a precedent for ClickNP-style decomposition applied to a specific RDMA workload. OpenURMA's contribution in this category is the first end-to-end UB data-path on commodity FPGA closing P&R at 322 MHz, not the FPGA-programmability story itself — our substrate is fixed-function once compiled.

Loss recovery and selective retransmission. IRN [32] is the canonical argument that an RDMA NIC should abandon Go-Back-N in favour of bitmap-tracked selective retransmission with per-packet ACKs, removing much of RoCE's reliance on PFC. MELO [30] sized the on-chip bookkeeping for scalable SR, using a shared bits pool to keep SACK metadata bounded regardless of connection count. FaSR [10] drives full-line-rate SR on the NIC with novel state-management schemes ad-

dressing the connection-vs-memory tension. DCP [28], the SIGCOMM'25 Best Student Paper Honorable Mention, presents an RTO-free, packet-LB-friendly hardware-offloadable SR design. LEFT [11] optimises packet reordering with a shared-bitmap pool and packet aggregation. OpenURMA's transport path implements both GBN and a selective `TPSACK`-driven retransmit slot (§3.4); we treat the above as algorithmic reference points, and a quantitative comparison under controlled loss injection (goodput-vs-loss-rate and retrans-buffer-occupancy curves) is left as follow-on work consistent with the SW-emu loss-injection gap flagged in §6.

Multi-tenant performance isolation. A second scalability axis — orthogonal to per-pair state — is isolation between cotenants sharing the same NIC. Justitia [43] provides software multi-tenancy via userspace pacing and rate-limiting. Husky [19] characterises how RNIC microarchitecture resources (MTT/MPT cache, processing units) can be exhausted by adversarial workloads, showing SR-IOV and HW TC do not isolate at the microarchitectural level. Harmonic [29] adds hardware-assisted isolation for public-cloud RDMA. OpenURMA does not currently address this axis; UB's per-host-pair connection model nevertheless removes one of the cache-pressure sources Husky exploits, since the cache no longer scales with application-pair count.

Multi-path and adaptive routing. MP-RDMA [31] first observed that letting RDMA WRITE/READ packets traverse multiple paths and arrive out-of-order, with bitmap-tracked completion, recovers substantial multi-path bandwidth without breaking the workloads that consume it. ConWeave [39] extends this into the switch with in-network reordering support for RoCE. STrack [21] is a hardware-offloaded multipath transport tuned for AI/ML clusters, combining adaptive load balancing, RTT-based CC, and SR-based loss recovery. Spectrum-X [34] adds adaptive-routing multi-path to commercial RoCE-on-Ethernet NICs. UB's TPG/spray [12] reuses the same architectural idea — spray flits across the path-diversity hash, recover order at the receiver — exposed as the default mode of the graded §7.3 ordering surface (UNO + NO); we implement TPG + spray in the `UB_TPG_Group` element (§3).

Software-controlled hardware transport. SCR / White-Boxing RDMA [44] is a complementary recent direction: it surfaces packet-granular software control over the BlueField-3 hardware transport via on-NIC Datapath Accelerators, enabling SW-defined CC, scheduling, and multi-path on top of vendor RoCE silicon. OpenURMA targets a different trade — the entire transport is RTL on FPGA, with no SW fast-path involvement — but the two are not mutually exclusive: a future revision could expose a similar SW-programmable surface on top of OpenURMA's open RTL.

Datacenter transports and congestion control. DCQCN [45] is the de-facto RoCE congestion control; we mirror it as the OpenRoCE baseline's CC. UB's C-AQM [12] is a queue-occupancy-based ECN signal; we model it as a SystemC + SW-emu element so the closed-loop behaviour is testable end-to-end without a physical switch.

Ordering semantics: total order vs. relaxed order. At the opposite end of the ordering design space from URMA, 1Pipe [26] provides a causally and totally ordered communication abstraction across a whole datacenter, with the goal of simplifying distributed transactions, state-machine replication, and distributed atomic operations. 1Pipe pays for the guarantee with in-network barrier aggregation at every switch and an extra $\sim$0.5–1 RTT of barrier-wait latency on each message, with reliable scatterings adding a further RTT for 2PC commit. URMA traffic is the wrong workload for that trade. The bulk of URMA verbs — ML training collectives, KV-cache / tensor transfers, parameter-server fan-in/fan-out, distributed checkpoint — are bandwidth-bound memory operations whose correctness contract is scoped to a single (Initiator, target Jetty) pair; cross-host total order is not part of the contract and paying for it costs real RTTs on the critical path. URMA's §7.3 graded surface (§2.2) is the specification-level descendant of MP-RDMA's relaxed-ordering insight discussed above: UNO + NO is the default fast path (no per-INI ordering at all, the regime where MP-RDMA's multi-path delivery is safe by construction), ROI / ROT / ROL graduate strictly more ordering as the workload demands it, and SO + Fence is the escape hatch for the rare verb that does need strict per-WR serialisation. 1Pipe-style total order remains a good fit for the higher-level distributed-transaction layer that may sit on top of URMA — not for the verb path itself, which is what this paper implements.

Adjacent fabrics. NVLink [33] and NVLink Fusion are NVIDIA-internal fabric protocols whose specification is partially open. CXL [3] is a load/store fabric over PCIe physical layers. The standard framing treats CXL and UB as different niches (intra-rack memory-coherence vs. scale-out DC-network); a more accurate framing is that they are different layers of the same architectural axis. CXL is the physical+link-layer mechanism for memory-semantic access at intra-chassis distances; UB's §8.3 Load/Store + §6.1 TP Bypass is the transport-layer mechanism for memory-semantic access at arbitrary cross-chassis distances. They are complementary rather than competing: a future system could route Load/Store to CXL while in-chassis and to UB across the rack, with the same ISA instruction reaching either substrate without an API change. UB's transport-layer contribution is precisely what permits the extension — a load/store semantic that does not terminate at the PCIe boundary the way CXL does.

System-layer τ-scaling. He [8] frames UB as the system-layer instance of a broader "τ scaling" principle: with geometric scaling's historical dividends flattening past 7 nm, the unifying optimisation target becomes the characteristic time constant τ at each layer, and the system-layer mechanism for compressing τ is to collapse the conversion-bound multi-protocol stack (PCIe→chassis fabric→Ethernet/IB→SW RDMA) into a single memory-semantic peer-to-peer fabric. He reports an empirical $\sim$500$\times$ end-to-end latency reduction for the UB ASIC, from $\sim$10 μs TCP/IP-class down to $\sim$100 ns. OpenURMA is the open-source artefact of the protocol layer in that argument: our 8-cycle ($\approx$25 ns) cold-path floor on commodity FPGA at 322 MHz is the same order as He's $\sim$100 ns ASIC target, and the mechanism — layer collapse via memory-semantic transport — is the one He identifies.

13. Conclusion

OpenURMA puts UB's three coupled architectural decisions — the Jetty/TP-Channel split, the graded §7.3 ordering surface, and the §8.3 Load/Store path through an on-chip-bus controller — into synthesisable RTL on commodity FPGA. 38 of 39 OpenURMA elements and all 21 OpenRoCE elements close 322 MHz Vivado P&R; the `UB_Jetty_Group` element follows the same II=2 template the four other ordering elements close at and is in-line as follow-on.

Three measurements, one architectural commitment. State. $O(N{+}M)$ vs $O(N{\cdot}M)$ is asymptotic, not constant: 5$\times$ at $(1,1)$, 4,855$\times$ at $(1024,1024)$, crossing the on-chip-BRAM-to-host-DRAM spill boundary between $N{\approx}23$ and $N{\approx}1024$. **Ordering.** 13 KLUT (10.6% of the design) and *zero* pipeline cycles on paths that do not request ordering; the cost is paid only on opt-in. **Latency.** 8 cycles ($\approx$25 ns) first-flit on the §8.3 path; $\approx$500 ns end-to-end CPU-to-remote-DRAM-to-CPU on a 64 B READ in the two-node simulator — 4.37$\times$ below the matched RoCE-DMA baseline (2186 ns). The gap is structural: the nine PCIe-bound cost components on the RoCE round trip (§8.5) are protocol consequences of disjoint CPU/NIC address spaces, and §8.3 removes the disjointness. **Price.** $\sim$14% of a U50's LUT budget; 46.6 ns of pipeline depth above the RC reference.

The three measurements compose as the introduction argued they would. The on-chip controller is feasible *only because* the layer split has bounded NIC context to $\sim$100 KB; the graded surface is cheap *only because* the per-Initiator counters already exist for it to gate; and the multi-path spray under UNO + NO is correctness-safe *only because* the two reorder buffers are split. OpenURMA's open RTL makes the commitment inspectable cell-by-cell, and its $\sim$14% U50 footprint says it is implementable on commodity silicon. The artefact is released as a substrate for follow-on work (multi-flit retransmission, alternative CC, multi-path scheduling) and as a controlled baseline against which Falcon, UEC, and SRNIC/StaR-derivative transports can be compared on the same infrastructure.

Code. `https://github.com/bojieli/OpenURMA`.

Acknowledgments. This work builds on the OpenClickNP toolchain (an open re-implementation of the ClickNP element model) and the published UB-Base-Specification 2.0.1.

References

[1] Mina Tahmasbi Arashloo, Alexey Lavrov, Manya Ghobadi, Jennifer Rexford, David Walker, and David Wentzlaff. Enabling programmable transport protocols in high-speed NICs. In *Proc. USENIX NSDI*, 2020.

[2] Brian F. Cooper, Adam Silberstein, Erwin Tam, Raghu Ramakrishnan, and Russell Sears. Benchmarking cloud serving systems with YCSB. In *Proc. ACM SoCC*, 2010. The Yahoo! Cloud Serving Benchmark; we use the YCSB-A 50/50 Get-Put Zipfian workload in §9.8.

[3] CXL Consortium. Compute Express Link (CXL) Specification 3.1. `https://www.computeexpresslink.org/`, 2024.

[4] Haggai Eran, Lior Zeno, Maroun Tork, Gabi Malka, and Mark Silberstein. NICA: An infrastructure for inline acceleration of network applications. In *Proc. USENIX ATC*, 2019.

[5] Daniel Firestone, Andrew Putnam, Sambhrama Mundkur, Derek Chiou, Alireza Dabagh, Mike Andrewartha, Hari Angepat, et al. Azure Accelerated Networking: SmartNICs in the public cloud. In *Proc. USENIX NSDI*, 2018.

[6] Adithya Gangidi, Rui Miao, Shengbao Zheng, Sai Jayesh Bondu, Guilherme Goes, Hany Morsy, Rohit Puri, Mohammad Riftadi, Ashmitha Jeevaraj Shetty, Jingyi Yang, Shuqiang Zhang, Mikel Jimenez Fernandez, Shashidhar Gandham, and Hongyi Zeng. RDMA over Ethernet for distributed training at meta scale. In *Proc. ACM SIGCOMM*, 2024.

[7] Dan Gibson, Hema Hariharan, Eric Lance, Moray McLaren, Behnam Montazeri, Arjun Singh, Stephen Wang, Hassan M. G. Wassel, Zhehua Wu, Sunghwan Yoo, Raghuraman Balasubramanian, Prashant Chandra, Michael Cutforth, Peter Cuy, David Decotigny, Rakesh Gautam, Alex Iriza, Milo M. K. Martin, Rick Roy, Zuowei Shen, Ming Tan, Ye Tang, Monica Wong-Chan, Joe Zbiciak, and Amin Vahdat. Aquila: A unified, low-latency fabric for datacenter networks. In *Proc. USENIX NSDI*, 2022.

[8] Tingbo He. A time scaling theory for multi-layer electronic systems. *ChinaXiv*, May 2026. chinarxiv-202605.00224. Perspective from Huawei Semiconductor: τ scaling as successor to geometric Moore's-Law scaling; positions Unified Bus as the system-layer τ reduction mechanism with end-to-end remote-access latency from $\sim$10s of μs (TCP/IP-class) to $\sim$100 ns.

[9] Hongjing Huang, Jie Zhang, Xuzheng Chen, Ziyu Song, Jiajun Qin, and Zeke Wang. SwCC: Software-programmable and per-packet congestion control in RDMA engine. In *Proc. USENIX ATC*, 2025.

[10] Peihao Huang, Guo Chen, Xin Zhang, Can Liu, Hongyu Wang, Huijun Shen, Ying Bian, Yuanwei Lu, Zhenyuan Ruan, Bojie Li, Jiansong Zhang, Yongfeng Liu, and Zhigang Chen. Fast and scalable selective retransmission for RDMA. In *Proc. IEEE INFOCOM*, 2025.

[11] Peihao Huang, Xin Zhang, Zhigang Chen, Can Liu, and Guo Chen. LEFT: Lightweight and fast packet reordering for RDMA. In *Proc. APNet*, 2024.

[12] Huawei Technologies. UB-base-specification 2.0.1. `https://www.unifiedbus.org/`, 2024. Unified Bus consortium specification, available from the consortium's documentation portal.

[13] Huawei Technologies. Ascend 950 NPU architecture white paper. Huawei vendor white paper, May 2026. Architectural disclosure for the Ascend 950PR and 950DT NPUs; first publicly documented silicon implementing the Unified Bus spec, with URMA (asynchronous Write/Read/Send/Atomic via Jetty) and UB Memory (synchronous Load/Store + AtomicStore/Load/Swap/CAS) exposed as distinct on-die paths, plus URMA-CTP / URMA-TP transport modes, UBoE 2$\times$400 Gbps, a hardware Collective Communication Unit (CCU), UB On-Chip Switch for in-die forwarding, and a UB super-node target of 8192 cards (cluster target $>$128K).

[14] Stephen Ibanez, Alex Mallery, Serhat Arslan, Theo Jepsen, Muhammad Shahbaz, Nick McKeown, and Changhoon Kim. NanoTransport: A low-latency, programmable transport layer for NICs. In *Proc. ACM SOSR*, 2021.

[15] Anuj Kalia, Michael Kaminsky, and David G. Andersen. Using RDMA efficiently for key-value services. In *Proc. ACM SIGCOMM*, 2014.

[16] Anuj Kalia, Michael Kaminsky, and David G. Andersen. FaSST: Fast, scalable and simple distributed transactions with two-sided (RDMA) datagram RPCs. In *Proc. USENIX OSDI*, 2016.

[17] Anuj Kalia, Michael Kaminsky, and David G. Andersen. Datacenter RPCs can be general and fast. In *Proc. USENIX NSDI*, 2019.

[18] Antoine Kaufmann, Tim Stamler, Simon Peter, Naveen Kr. Sharma, Arvind Krishnamurthy, and Thomas Anderson. TAS: TCP acceleration as an OS service. In *Proc. EuroSys*, 2019. Reports detailed PCIe-class transaction latency decompositions used as parameter references in §8.

[19] Xinhao Kong, Jingrong Chen, Wei Bai, Yechen Xu, Mahmoud Elhaddad, Shachar Raindel, Jitendra Padhye, Alvin R. Lebeck, and Danyang Zhuo. Understanding RDMA microarchitecture resources for performance isolation. In *Proc. USENIX NSDI*, 2023.

[20] Xinhao Kong, Yibo Zhu, Huaping Zhou, Zhuo Jiang, Jianxi Ye, Chuanxiong Guo, and Danyang Zhuo. Collie: Finding performance anomalies in RDMA subsystems. In *Proc. USENIX NSDI*, 2022.

[21] Yanfang Le, Rong Pan, Peter Newman, Jeremias Blendin, Abdul Kabbani, Vipin Jain, Raghava Sivaramu, and Francis Matus. STrack: A reliable multipath transport for AI/ML clusters. arXiv:2407.15266, 2024.

[22] Bojie Li. OpenClickNP: a clean-room reimplementation of ClickNP on Alveo U50. `https://github.com/bojieli/OpenClickNP`, 2025–2026.

[23] Bojie Li, Zhenyuan Ruan, Wencong Xiao, Yuanwei Lu, Yongqiang Xiong, Andrew Putnam, Enhong Chen, and Lintao Zhang. KV-Direct: High-performance in-memory key-value store with programmable NIC. In *Proc. ACM SOSP*, 2017.

[24] Bojie Li, Kun Tan, Layong Larry Luo, Yanqing Peng, Renqian Luo, Ningyi Xu, Yongqiang Xiong, Peng Cheng, and Enhong Chen. ClickNP: Highly flexible and high-performance network processing with reconfigurable hardware. In *Proc. ACM SIGCOMM*, 2016.

[25] Bojie Li, Zhilong Xiang, Xiang Wang, Hongru Jonathan Zhou, and Kun Tan. FastWake: Revisiting host network stack for interrupt-mode RDMA. In *Proc. APNet*, 2023.

[26] Bojie Li, Gefei Zuo, Wei Bai, and Lintao Zhang. 1Pipe: Scalable total order communication in data center networks. In *Proc. ACM SIGCOMM*, 2021.

[27] Qiang Li, Yixiao Gao, Xiaoliang Wang, Haonan Qiu, Yanfang Le, Derui Liu, Qiao Xiang, Fei Feng, Peng Zhang, Bo Li, Jianbo Dong, Lingbo Tang, Hongqiang Harry Liu, Shaozong Liu, Weijie Li, Rui Miao, Yaohui Wu, Zhiwu Wu, Chao Han, Lei Yan, Zheng Cao, Zhongjie Wu, Chen Tian, Guihai Chen, Dennis Cai, Jinbo Wu, Jiaji Zhu, Jiesheng Wu, and Jiwu Shu. Flor: An open high performance RDMA framework over heterogeneous RNICs. In *Proc. USENIX OSDI*, 2023.

[28] Wenxue Li, Xiangzhou Liu, Yunxuan Zhang, Zihao Wang, Wei Gu, Tao Qian, Gaoxiong Zeng, Shoushou Ren, Xinyang Huang, Zhenghang Ren, Bowen Liu, Junxue Zhang, Kai Chen, and Bingyang Liu. Revisiting RDMA reliability for lossy fabrics. In *Proc. ACM SIGCOMM*, 2025. Best Student Paper, Honorable Mention.

[29] Jiaqi Lou, Xinhao Kong, Jinghan Huang, Wei Bai, Nam Sung Kim, and Danyang Zhuo. Harmonic: Hardware-assisted RDMA performance isolation for public clouds. In *Proc. USENIX NSDI*, 2024.

[30] Yuanwei Lu, Guo Chen, Bojie Li, Kun Tan, Yongqiang Xiong, Peng Cheng, Jiansong Zhang, Enhong Chen, and Thomas Moscibroda. Memory efficient loss recovery for hardware-based transport in datacenter. In *Proc. APNet*, 2017.

[31] Yuanwei Lu, Guo Chen, Bojie Li, Kun Tan, Yongqiang Xiong, Peng Cheng, Jiansong Zhang, Enhong Chen, and Thomas Moscibroda. Multi-Path transport for RDMA in datacenters. In *Proc. USENIX NSDI*, 2018.

[32] Radhika Mittal, Alexander Shpiner, Aurojit Panda, Eitan Zahavi, Arvind Krishnamurthy, Sylvia Ratnasamy, and Scott Shenker. Revisiting network support for RDMA. In *Proc. ACM SIGCOMM*, 2018.

[33] NVIDIA Corporation. NVLink: A high-bandwidth inter-GPU interconnect. Vendor whitepaper, 2014–2025. Successive generations of the NVLink fabric are described in the NVIDIA whitepaper series.

[34] NVIDIA Networking. NVIDIA Spectrum-X: Adaptive routing and telemetry-based congestion control for AI networks. NVIDIA technical brief, 2024. Vendor description of multi-path adaptive-routing delivery over Spectrum-4 / BlueField-3 NICs; the closest commercially-deployed point of comparison to UB's TPG multi-path scheme.

[35] Sebastian Ramos and Torsten Hoefler. Designing high-performance, low-latency multi-cluster communication on modern InfiniBand networks. In *Proc. ACM HPDC*, 2023. Reports ConnectX-7 PCIe round-trip latencies in the $\sim$300–500 ns range; we use this as the parameterised PCIe RTT in §8.

[36] David Sidler, Zeke Wang, Monica Chiosa, Amit Kulkarni, and Gustavo Alonso. StRoM: Smart remote memory. In *Proc. EuroSys*, 2020.

[37] Arjun Singhvi, Aditya Akella, Dan Gibson, Thomas F. Wenisch, Monica Wong-Chan, Sean Clark, Milo M. K. Martin, Moray McLaren, Prashant Chandra, Rob Cauble, et al. 1RMA: Re-envisioning remote memory access for multi-tenant datacenters. In *Proc. ACM SIGCOMM*, 2020.

[38] Arjun Singhvi, Nandita Dukkipati, Prashant Chandra, Hassan M. G. Wassel, Naveen Kr. Sharma, Anthony Rebello, Henry Schuh, Praveen Kumar, Behnam Montazeri, Neelesh Bansod, Sarin Thomas, Inho Cho, Hyojeong Lee Seibert, Baijun Wu, Rui Yang, Yuliang Li, Kai Huang, Qianwen Yin, Abhishek Agarwal, Srinivas Vaduvatha, Weihuang Wang, Masoud Moshref, Tao Ji, David Wetherall, and Amin Vahdat. Falcon: A reliable, low latency hardware transport. In *Proc. ACM SIGCOMM*, 2025. Specification contributed to OCP 2024.

[39] Cha Hwan Song, Xin Zhe Khooi, Raj Joshi, Inho Choi, Jialin Li, and Mun Choon Chan. Network load balancing with in-network reordering support for RDMA. In *Proc. ACM SIGCOMM*, 2023.

[40] Ultra Ethernet Consortium. Ultra Ethernet specification 1.0. Industry specification, 2025. Released June 2025 under Linux Foundation JDF; `https://ultraethernet.org/`.

[41] Xizheng Wang, Guo Chen, Xijin Yin, Huichen Dai, Bojie Li, Binzhang Fu, and Kun Tan. StaR: Breaking the scalability limit for RDMA. In *Proc. IEEE ICNP*, 2021.

[42] Zilong Wang, Layong Luo, Qingsong Ning, Chaoliang Zeng, Wenxue Li, Xinchen Wan, Peng Xie, Tao Feng, Ke Cheng, Xiongfei Geng, Tianhao Wang, Weicheng Ling, Kejia Huo, Pingbo An, Kui Ji, Shideng Zhang, Bin Xu, Ruiqing Feng, Tao Ding, Kai Chen, and Chuanxiong Guo. SRNIC: A scalable architecture for RDMA NICs. In *Proc. USENIX NSDI*, 2023.

[43] Yiwen Zhang, Yue Tan, Brent Stephens, and Mosharaf Chowdhury. Justitia: Software multi-tenancy in hardware kernel-bypass networks. In *Proc. USENIX NSDI*, 2022.

[44] Chenxingyu Zhao, Jaehong Min, Ming Liu, and Arvind Krishnamurthy. White-boxing RDMA with packet-granular software control. In *Proc. USENIX NSDI*, 2025.

[45] Yibo Zhu, Haggai Eran, Daniel Firestone, Chuanxiong Guo, Marina Lipshteyn, Yehonatan Liron, Jitendra Padhye, Shachar Raindel, Mohamad Haj Yahia, and Ming Zhang. Congestion control for Large-Scale RDMA deployments. In *Proc. ACM SIGCOMM*, 2015.